

U.S. Department of Agriculture



Idaho Panhandle National Forests

Tree Planting and Associated Services IDIQ

Performance of Work Statement

Idaho Panhandle National Forests

North Zone-Bonnars Ferry, and Sandpoint Ranger Districts

Bid-Item 25 Tree Planting

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Performance Work Statement

Project Title

Idaho Panhandle National Forests Planting IDIQ

1.0 General Information

Background and Purpose

The Idaho Panhandle National Forests needs planting and related services on lands administered within its boundaries. Planting and related services are needed on lands that have been disturbed through fire, insect or diseases, harvest, or from other causes and where land and resource management plans indicate forested cover is desired after these disturbances.

Scope of Contract

The Contractor shall furnish everything necessary, including all labor, equipment, supervision, transportation (i.e., crews), supplies (except those as designated as Government furnished), and incidentals to perform all work necessary on the areas specified in task orders issued under this contract. The scope of this contract includes, but is not limited to, tree transport, tree care, handling, and planting, tree preparation (if needed for bare root stock), natural and artificial shading treatments, animal damage control treatments (i.e., protective netting), and related services.

Location and Description

Task orders may be issued on any of the Zones listed in the Schedule of Items. Information on each Zone within the Idaho Panhandle:

North Zone (Bonners Ferry, Sandpoint and Priest Lake Ranger Districts).

Central Zone (Coeur d'Alene and Silver Valley).

South Zone (St Maries and Avery).

Specific locations, boundary identification, maps, and data for individual planting units or treatment areas will be provided on task orders.

Stock Types

There is a combination of stock types used throughout the Idaho Panhandle NF both bareroot and container- grown seedlings. The entire list of stock types that could be used throughout the life of the contract include, but are not limited to, the following:

Stock Type Name	Container Volume/Bareroot Specifications
1-0 Bareroot	See section 3.0 - <i>Government Furnished Property</i> in task orders for seedling specifications
2-0 Bareroot	See section 3.0 - <i>Government Furnished Property</i> in task orders for seedling specifications
Plug ½	combination bareroot and container
Container-grown 160	5.5 cubic inch
Container-grown 112	6.0 cubic inch
Container-grown 91	7.9 cubic inch
Pots	1 gallon

Planting Seasons

Forests in the Intermountain Region generally plant during two seasons, spring and fall. Spring planting seasons run from approximately April 15-June 15 and fall planting seasons run from approximately

September 15-October 15. All dates are approximate and depend on weather and/or local site conditions. Task orders may be issued for one or more planting seasons within any given year.

Seedling survival and growth is dependent on proper soil and moisture conditions; these conditions may prevail for only a short period of time. It may be necessary to plant a portion of a unit and return when ground conditions are suitable to plant the remaining area.

Safety

There may be snags and/or burned trees present in any or all the units to be planted under this contract. Some of the units to be planted may have been burned by wildfire. There is potential for blowdown or falling debris especially in windy conditions. See section 9.0 – *Deliveries or Performance – Safety Plan* regarding providing a Safety Plan.

2.0 Definitions

Acceptable Existing Seedling. Acceptable existing trees are defined as: Conifer – six inches or greater in height that appear to be growing well. Aspen – 20 or more stems two feet in height.

Acceptable Existing Tree. Any live tree meeting diameter specifications in the Task Order Detailed Information Sheets will define these requirements; a healthy tree with good form, vigor and foliage coloration, and shall have at least the top 1/4 of the tree in live green foliage. Such a tree shall be free of disease (i.e., dwarf mistletoe brooming), and severe physical damage (i.e., broken out top, three or more forks, leaning more than 30 degrees from vertical, severe crook or severe cambial damage, bark flaking caused by woodpeckers).

Acceptable Quality Level (AQL). The allowable leeway or variance from a standard before the government will reject the specific service. An AQL does not say that the contractor may knowingly offer defective service. It implies that the government recognizes that defective performance sometimes happens unintentionally. If the percent of defective performance does not exceed the AQL, the service will not be rejected by the government. The contractor, however, must re-perform the defective service when possible.

Access Point. The point identified for vehicle access to planting areas. Contractor vehicles will be able to drive to this point. Government furnished property, which includes the planting stock, will NOT be delivered to an access point UNLESS the same location has been designated as a delivery point.

Animal Damage Protection. Methods for protecting the seedlings from animal damage, particularly big game. May include Vexar-type tubing or flexible netting.

Artificial Shade. Materials not found on the planting site used to shade the tree.

Bareroot Trees. Seedling grown in a field and then lifted from the soil when dormant, removing soil from the roots. Roots should be protected from drying out.

Clean Water and Buckets/Tubs. Water and buckets/tubs used for tree dipping that are free of contaminants that could injure the trees. Water is initially suitable for human consumption.

Clearing. The removal of surface debris in and around the planting hole, exposing mineral soil. Debris includes duff, litter, rotten or charred wood, moss, loose rock, ashes, snow, and surface frost.

Removing this material avoids debris from falling into the planting hole.

Container-grown Tree. Seedling grown in a receptacle containing soil and nutrients. When the container seedlings are extracted from the receptacle, the soil stays intact around the roots. The soil and roots together are referred to as the container plug; the soil is the container media.

Contract Time. See Period of Performance

Contracting Officer (CO). Individual with authority to bind the government with their signature on the contract.

Contracting Officer Representative (COR). Individual(s) designated to act on the behalf of the CO within the delegation given them in writing.

Cotyledon Scar. The location where the seedcoat had been attached to the seedling. This marks the point where the hypocotyl, which grows upward and turns into the stem and the radicle, which becomes point of differentiation between the root system and the shoot.

Delivery Point. The point identified for delivery of trees and other supplies. If there are multiple points for delivery of various items, it will be so noted. May be at the planting unit, at the end of a walk-in, district cooler, etc. A delivery point may or may not be the same location as an access point. Wherever supplies are to be delivered is considered a delivery point. An access point is only a location to which a Contractor can drive vehicles to a planting unit.

Diameter at Breast Height. The diameter of the stem of a tree measured outside bark at breast height (4.5 feet or 1.37 m) from the ground. On sloping ground, the measure is taken from the uphill side.

Drip Line. The line extending vertically from the exterior edge of a tree's live crown to the ground.

Excess Trees. Trees that are tallied on the Planting Inspection Sheet that exceed the maximum allowable trees to be planted. These trees are considered wasted trees.

Foreign Material (in planting hole). Large rocks, sticks, litter or similar material not part of the surrounding soil substrate that is in the planting hole after the tree is planted.

Government Technical Monitor (GTM). Individual(s) designated by the CO to assist the COR in monitoring a contractor's performance.

In-box Temperature. The temperature near the center of the box of trees measured with a probe thermometer or similar instrument. This is the temperature of the seedlings while in the shipping box.

Lateral Roots. The horizontal roots extending from the main vertical (tap) root.

MTR or M Trees. Abbreviation for measurement of 1,000 trees.

Microsite. A plantable spot favorable to seedling survival and growth. Examples of favorable microsites are stumps, down logs, large woody slash, snags, and standing cull trees that provide protection from sun, wind, animals, and other damaging agents.

Moist Soil (in the planting hole). The soil in the planting hole is considered "moist" if it is similar in

moisture to the undisturbed soil next to the planting hole at similar depth.

Pay or Sub Item. A unit or group of units listed under one bid item on the Schedule of Items.

Period of Performance (also Performance Period or Contract Time). All calendar days allowed for completion of contract work. Includes all days of the week including weekends and holidays.

Plantable Spot. An area from which vegetation, ash, duff, and debris have been, or can be, removed and seedling can be planted according to all specifications contained herein. In planting inspections, a plantable spot is also any areas where the Contractor has planted a tree.

Planting Inspection. The standard planting inspection procedure provided in *Section 8.0*. It includes the description of planting violations that result in unsatisfactorily planted trees, plot sizes, calculations for determining planting quality based on trees planted, and similar information.

Protection. Natural and artificial materials that act as an obstacle protecting the tree from animals or other injurious agents. The protection material and effort required by the planter to protect the tree varies depending on the site conditions of each unit.

Quality Assurance (QA). Those actions taken by the Government to check goods or services to determine that they meet the requirements of the Performance Work Statement (PWS).

Quality Assurance Surveillance Plan (QASP). An organized written document used for quality assurance surveillance. The document contains information about how the government will assess performance, checklists, and may contain decision tables.

Quality Control/Quality Control Plan (QCP). Those actions taken by a contractor to control the production of goods or services so that they will meet the requirements of the PWS. The QCP is the Contractor's plan for self-monitoring work prior to Government inspection and acceptance.

Random Sampling. A method of looking at a few individual items in a lot to determine the quality of that lot against a standard.

Roads. Primary access roads will provide the closest access possible to units

Root Collar. The original ground line of the tree before planting is the soil surface level when the tree was in the nursery beds or in the container receptacle. The ground line is represented by the "root collar". Above this point is stem tissue, below this point is root tissue.

Root Length. The root length measured from the original ground line of the tree to the end of the main vertical or tap root. For container-grown trees, this is the length of the plug.

Satisfactory Planted Seedling. A seedling planted in full compliance with all planting specifications.

Scalping. The removal of live vegetation, including roots, in and around the planting hole to reduce competition to the planting tree. Vegetation is removed down to and exposing mineral soil.

(Acceptable) Service. A job performed to the standard and within the AQL. The contractor must do the specific job, and meet the standard, and meet the acceptable quality level before one can say that performance has been acceptable and that he/she should be paid.

Shade. Natural and artificial materials that protect the tree from the sun's heat and wind.

Standard. An acknowledged measure of comparison.

Stationary Shade Material. Natural material used to shade or protect the tree that is not moveable by the planter. Natural material includes items such as stumps, logs, dead brush, rocks, or terrain features used to provide the required protection for a tree. Stationary material generally offers the most shade and protection for seedlings from sun and animal trampling.

Top Height. The length from the original ground line of the tree to the top of the terminal leader; does not include needles extending above the leader.

Transportable Shade Material. Natural material such as wood debris that can be moved to the planting spot to shade the tree from the sun. Although the planter is expected to put reasonable effort into moving shade material, it should be able to be done without putting down the planting tool.

Trees in Possession of the Contractor. The time after trees have been issued to the Contractor. During this time, the Contractor is responsible for all aspects of tree care and accountable for tree quantities.

Unplantable Spot. An area within the specified spacing limits in which it is not possible to plant a seedling according to specifications and no tree has been planted. One unplantable spot is allowed for each single unplantable area equal in size to the average specified spacing. Example: if a single unplantable area of 64 feet exists on an 8-by 8 foot spacing, one unplantable spot is recognized.

In hoedad planting, a spot is considered unplantable if a suitable spot for planting a tree cannot be found after three attempts of throwing the hoe, and if the hole cannot be opened at the spot with five swings or less.

In auger planting, the spot is considered unplantable if a suitable plantable spot cannot be found after three attempts at clearing or scalping, and then after three attempts to open the hole in the spot with the auger fail.

Unsatisfactory Planted Seedling. A planted seedling which fails to meet one or more of the specifications for a satisfactorily planted seedling.

Walk-in Units. Some units cannot be accessed from open forest roads and require the Contractor to walk; vehicle access is to the 'access point' identified on the project map. If the access point and the delivery point are the same location, the Contractor shall transport trees and materials from that location to the unit using the method of his/her choice consistent with restrictions on motorized vehicles or other transport method noted on the Detailed Information Sheet in task orders.

Wasted Material. Government furnished property that is lost, dropped, or damaged by the Contractor, or which is installed on trees not in accordance with the requirements of the contract. This may include, but is not limited to, installing shade on the wrong species, or netting on the wrong tree(s).

Wasted Seedlings. Seedlings which are lost, damaged, destroyed, or handled contrary to the specifications listed under *Section 4.0 – Tree Handling and Care*. Planted trees that exceed the

maximum number of trees in the planting inspection procedure are also considered wasted seedlings.

3.0 Government Furnished Property

Item	Description	Where Delivered
Bare Root Trees	Length of roots and top heights will be defined in task orders. At least 93% of the trees will meet these specifications.	As specified in task orders
Container-Grown Trees	Length of container plug and top heights will be defined in task orders. At least 93% of the trees will meet these specifications.	As specified in task orders
Tree dipping materials	Buckets/tubs, Water for dipping	As specified in task orders
Forest Service Locks	Forest Service locks on gates will be unlocked by Government so that government and Contractors have access to planting units.	N/A

4.0 Contractor Requirements/Tasks

Access

Road Access

Planting units are accessible by four-wheel drive vehicles equipped for mountain driving, except where task orders specify walk-in units. The Government will provide passage to access points noted on task order maps for units either accessible from roads or from walk-ins. After initial clearing of the access roads by the Government, the Contractor shall clear any subsequent obstacles that block passage to access points.

Walk-in Units

These units cannot be accessed with vehicles from open forest roads. The Contractor shall transport trees and any shading or ADC materials (if required on task orders) from access points noted on task order maps to units. Transport methods shall not violate Forest-specific restrictions on off-road use, unless specifically excepted in task orders or by the COR.

Locked Gates

In instances where units are behind locked gates, the Contractor shall close and lock gates after passing through. The Contractor shall be subject to all restrictions of administrative use. If the gate has a double lock, ensure that when relocking gates, the non-Contractor accessible lock can be opened properly to allow others to pass through. The Contractor shall drive from the locked gate to access points with only the minimum number of vehicles necessary for efficient transport of the crew and trees. No firearms are permitted behind locked gates.

Tree Transport Within Unit

The Contractor may transport trees from access points to any areas within units consistent with *Section 4.0-Tree Handling and Care* and ensuring that Forest-specific restrictions on off-road use are not violated, unless specifically excepted in task orders or by the COR.

Orders, Delivery, and Returns

Government-furnished property shall be inventoried upon Contractor pickup and if authorized, return.

Refer to *Exhibit 12 – Tree Tracking Sheet*.

Seedlings

The Contractor shall order trees at least 24 hours in advance of desired delivery. This time period is necessary to allow the Government time to prepare trees for planting and for acclimatization if needed.

The Contractor shall plant all trees ordered and delivered to the planting unit within 12 hours of delivery, unless otherwise approved by the COR. Trees shall not be returned to the Government for cooler storage unless approved by the COR. When the COR determines that conditions are unsuitable for planting, the Contractor shall return all unplanted trees to the delivery point with no penalty for delay in planting the delivered trees. If returning trees to storage is authorized, the trees shall be returned to the shipping boxes and marked to indicate the seed lot or other information as specified by the COR. These trees will not be counted as wasted trees.

Any trees not planted out and returned without approval by the COR shall be considered wasted trees.

Artificial Shade and ADC Materials

The Contractor shall order materials for artificial shade or animal damage control treatments at least 24 hours in advance of desired delivery. Materials will be delivered to access or delivery points as identified in task orders.

Return of Government Furnished Property

The Contractor shall return unused Government supplies, unplanted trees, all packaging, wrapping, shade, and ADC materials to access or delivery points at the end of each planting day, as specified in task orders.

Tree Handling and Care

The Contractor shall be responsible for the care of trees from the delivery point and throughout the time that trees are in possession of the Contractor to assure they are properly cared for and not damaged or exposed to damaging conditions. Trees not properly cared for may be considered wasted trees. The Contractor shall be aware of environmental hazards which affect tree survival; he/she shall alter crew work schedules and tree handling practices to allow planting whenever possible. Care shall be taken to adhere to the following requirements any time trees are in the Contractor's possession, with noncompliance to these requirements grounds for damage payments or for termination of the contract:

- The Contractor shall take precautions to keep roots moist during the entire planting operation.
- Trees shall be protected from drying, heating, smothering, freezing, crushing, bending, drowning, abrasion, rapid temperature fluctuations, or shall not come into contact with injurious substances such as petroleum products or petroleum product residues.
- Seedlings removed from cold storage facilities shall not stand or lay in water or snow or be covered directly with snow.
- Shipping containers containing trees shall be opened only in full shade and shall not be exposed to direct sunlight. Shipping containers shall not be tossed or dropped. Punctured or torn shipping containers or liners shall be promptly resealed. Shipping containers shall be separated to provide free air movement. Trees may be placed on racks or inside boxes which will support rack or box weight and prevent crushing.
- The temperature in shipping box shall be maintained at 45 degrees F, or similar to the temperature in the box at receipt, whichever is higher.

- Trees shall not be removed from shipping containers until needed for planting preparation.
- Trees are to be planted or prepared without further root or top pruning or culling. If pruning or culling appears necessary, or if mold, dry roots, or evidence of injury or drying is seen, the condition shall immediately be reported to the COR.
- Trees that are frozen shall not be handled until completely thawed. They shall be thawed in accordance with direction from the COR.
- Trees are to be kept in the baggies they were packed in when placed in planting bags.
- Trees in planting bags shall only have tops exposed.
- Trees carried in planting bags shall not exceed the amount that can be carried and removed without injury, or which can be planted before critical heating or drying occur.
- Trees shall not be removed from planting bags until immediately before planting. They shall be handled carefully to avoid damage to roots and buds. They shall be gently removed one at a time to prevent injury and quickly inserted into the prepared planting hole. All trees in a planter's bag should be out planted prior to taking breaks.
- Trees placed in bags will be planted and not returned to storage, unless approved by the COR as described in *Section 4.0 – Orders, Delivery, and Returns*.

Caches

Seedling cache sites/temporary storage shall be allowed within or near the planting units. These sites are intended for staging seedlings on the same day that they are to be planted. The COR shall approve all seedling caches/temporary storage sites. COR approved seedling protection tarps (**to be furnished by the Government**) will be utilized to protect the seedlings within the cache. Tarps shall be white on one side and have a reflective surface on the other.

Equipment

The Contractor shall provide all transport for seedlings, planting tools, bags, and other necessary equipment to perform planting and shading/ADC treatments and quality control. Required planting tools will be included in task order Detailed Information Sheets.

Tree Transport to Planting Units

Special care shall be taken when transporting seedlings to access points (follow *Section 4.0 – Tree Handling and Care* section above). Contractor transport of planting stock shall be by either fully enclosed trailers or pickups with fully enclosed canopies. Trees may not be carried with petroleum products, petroleum powered tools, or in a petroleum contaminated enclosure.

Hand Tools

Tools shall be capable of opening a hole that is deep and wide enough to accommodate a 9-inch plug plus a minimum 1-inch burying depth, unless otherwise specified in task orders. Hand tools used shall be capable of chopping and cutting live or dead vegetation and adequate to create minimum or larger scalp sizes listed in task order Detailed Information Sheets.

Power Augers

Power augers, if specified on task orders, shall be lubricated with vegetable oils or similar substances not injurious to trees and shall not leak gas.

Planting Bags

Planting bags shall be a minimum of 15 inches deep, free of defects, light in color, of porous material

(i.e., canvas), and shall not retain water or heat. Bags may have a shield of waterproof material attached to one exterior side to protect the planters from water seepage. Planting bags must not be transported with or come into contact to petroleum-based products.

Tree Preparation

Dipping – Bareroot Stock Only

Container-grown trees will not be dipped or wrapped.

Water Dip

The Contractor shall dip the roots of bareroot trees in clean water at the planting site, prior to placing trees in the planter's bag. Dipping in water is intended to increase root moisture. Roots shall be fully wet, but soil particles shall not be washed off the roots during the dipping process. Bareroot stock shall be removed from the shipping boxes and immediately dipped in water. Tree roots shall not be exposed to air for more than 20 seconds. All tree sorting and packing activities shall be done in shaded areas and protected from the wind. The shipping bags shall be kept closed at all times except when trees are being removed.

All dipping shall be done under the direct supervision of the Government. The planting crew will be required to dip as an entire group during the course of the day. Tree planters will not be allowed to return to the dipping site in a random fashion to dip trees on an individual basis. With approval from the CO and oversight from CORs or GTMs, the Contractor may send individuals to the dip site to get trees for the rest of the planting crew (tree runners or carriers).

Water for Dipping

The Contractor will haul water necessary for dipping operations. Water can be brought with or acquired from a location agreed to by the COR. Contractor shall utilize at least one dipping tub per four planters. Dipping tubs shall be a minimum of 14 inches deep. Bareroot trees shall be dipped for not less than 30 seconds or more than four minutes. For ponderosa pine, trees shall be dipped in water only long enough to moisten the tree roots (ten seconds or less). After dipping, Sontara/Kimtex tree planting liners shall be moistened to maintain root moisture. Trees shall not be forcibly packed into water tubs or tree planting bags.

Planting Bag Liners

Planting bags will have insulated liners constructed from ½ inch thick closed cell polyurethane, Sontara or Kimtex jelly-roll material surrounded by polyester mesh, or another moisture-retaining liner approved by the CO. Planting bag liners shall be kept saturated with water, but all excess water shall be drained prior to putting trees in the planting bag.

Inventory

The tree inventory on each nursery box shall serve as the official tree count for dipping operations. No additional counting of trees will be allowed to minimize seedling exposure time outside of boxes of planting bags.

Tree Planting

Scalping

Each planting spot shall be free of live vegetation, including roots, to reduce competition from competing vegetation. The Contractor shall remove the vegetation to the maximum sizes specified in Detailed Information Sheets. The Contractor shall not destroy or remove any existing tree seedling unless otherwise specified by the COR. Drainage of scalps is not required.

Clearing

Each planting spot shall be free of surface debris to avoid foreign material from falling into the hole. All debris down to the soil surface in areas according to the maximum sizes specified on task order Detailed Information Sheets shall be removed by the contractor. Debris to be removed includes duff, litter, dry soil, rotten or charred wood, moss, loose rock, ashes, snow and surface frost.

Tree Placement, Filling, and Firming

To ensure the greatest probability of tree survival and growth, trees shall be planted according to examples shown in Exhibit 02 and according to the following:

- Should the initial attempt to open a planting hole fail because of subsurface rocks, roots, or other obstacles, a second effort shall be made in a slightly different part of the planting spot. If this effort fails for similar reasons, a third attempt shall be made in another part of the planting spot. If the third effort fails, spot shall be abandoned as unplantable.
- All additional and auxiliary holes created during the planting operations that are within 18 inches of the planted tree shall be filled-in or collapsed.
- Planting holes shall be located as near to the center of the cleared and scalped area, except as required to meet artificial shade; stationary material (i.e roots, rocks) or ADC installation and shall be oriented at an angle between perpendicular to the slope and vertical.
- Trees shall be planted in an open hole, deep and wide enough to fully accommodate the roots of the tree being planted. The tree roots shall be in their natural configuration and not be wedged, jammed, spiraled, or twisted into the hole.
- The hole shall be filled and firmed with the planter's hands, using moist soil taken from the planting hole so the tree is firmly in the soil, there are no air pockets, and the tree is firmly secured in the ground. Firming the soil around the tree shall be done in a manner to ensure no damage is done to the tree or root system. There shall be no material in the hole which is foreign to the hole such as dry solids, ash, organic matter, and rock.
- After soil is packed around the seedlings, it shall be smoothed out to the level of the surrounding mineral soil surface, cupping will not be allowed.
- When bareroot trees are planted with lateral roots exceeding the maximum length, the long lateral roots shall be placed in the bottom of the planting hole in a manner that does not affect the placement of the vertical root or other lateral roots. Trees with long lateral roots planted in this manner will not be subjected to a root configuration violation during inspection.
- Bareroot trees shall be planted deep enough so that after filling and firming, the soil surface is at, or up to one inch above, the original ground line of the tree with no roots exposed and no branches buried. The root system shall extend downward and be aligned with the axis of the planting hole.
- Container-grown trees shall be planted so that after filling and firming, there is ½ to 1 inch of soil

on top of the container plug. The cotyledon scar should be level with the surrounding soil surfaces to seal the plug into the ground. The Contractor shall take care to keep the container growing media intact. Where media falls away from the roots, the tree shall be planted in accordance with the Bareroot trees guidance listed in the previous bullet.

- After planting, the tree stem shall be erect and oriented between vertical and perpendicular to the slope. Trees or tree branches shall not be weighted down with mud or debris. Tree branches shall not be firmed into the soil. There shall be no excessive depressions on the ground surface that may collapse on the tree hole and the tree shall not be in contact with the dirt bank. The tree shall be free from damage.
- Once a tree has been planted, the tree shall not be uprooted and replanted.
-
- *Tree Species or Stock Type Mix*

The species and stock types to be planted along with the distribution of each will be indicated on task order Detailed Information Sheets. The Contractor shall maintain a uniform mixture of species throughout the unit unless otherwise described on Detailed Information Sheets.

Acceptable Established Seedlings and Acceptable Existing Trees

A tree shall not be planted in direct competition with established seedlings, other planted trees or under the immediate influence of acceptable existing trees. Task order Detailed Information Sheets will define these requirements. Trees not meeting acceptable existing seedling or tree definition shall be disregarded and a seedling planting according to task order specifications.

Seedlings shall not be planted outside the original forest canopy. Natural openings are considered to be outside the original forest canopy and shall not be planted.

Shade Requirements (S.1)

Planting in favorable microsites protects seedlings from potentially harmful conditions and improves the probability of survival. Trees shall be planted consistent with the shade requirement for each planting unit as specified in task order Detailed Information Sheets. The type of shade available and the effort required by the planter to shade seedlings varies depending on the site conditions of each unit. Shade material includes downed logs, stumps, and standing snags meeting minimum specifications.

When shade is required, each tree shall be shaded at the ground level from the afternoon sun. Trees shall be planted on the north, northeast, or east side of the shade material (refer to Exhibit 03). Each tree shall be as close to the shade material as possible (without touching), and within $\frac{1}{2}$ the distance equal to the height of the shade material (for stumps, downed logs and similar) or $\frac{1}{2}$ the distance equal to the width of the shade material in the case of standing snags. Exceptions will be noted on task order Detailed Information Sheets.

A stationary shade source may be used as part of the scalp.

When transportable shade material is required, it must be at least 3 inches in diameter and be available within $\frac{1}{2}$ of the distance to the next plantable spot, unless otherwise specified in task order Detailed Information Sheets. Do not place shading material where it can roll on planted trees.

S.1.1-Stationary Material Only

Tree shall be planted next to stationary material when it is available. Planting spot selection takes precedence over spacing requirements. When stationary material is not available, no tree is planted.

S.1.2-Plant Existing Shade Only

Planting spot selection takes precedence over spacing requirements. Trees will be planted (regardless of spacing) next to stationary shade sources, listed in order of preference with the most favored first, on task order Detailed Information Sheets. Planting spot selection will move down the list if higher priority stationary shade sites don't exist. If no priority sites exist, seedlings are planted to achieve spacings as described on task order Detailed Information Sheets. Refer to Exhibit 05 for an example of this method.

S.1.3-Stationary Material First, Then Transportable Shade

A tree shall be planted next to stationary material when it is available. When stationary material does not exist in the spacing allowance, shade shall be transported to the seedlings. When neither stationary nor transportable material exists, no tree shall be planted.

S.1.4-Stationary Material First, Transportable Shade Second, Plant in Open Last

A tree shall be planted next to stationary material when it is available. If stationary material does not exist in the spacing allowance, shade shall be transported to the tree. When neither stationary nor transportable material exists, the tree may be planted in the open.

S.1.5-Stationary Material or Transportable Shade (Planter Discretion)

A tree may be planted next to either stationary material or transportable shade at the planter's discretion. When neither stationary nor transportable material is available, no tree is planted.

S.1.6-Stationary Material or Transportable Shade (Planter Discretion), then Plant in Open

A tree may be planted next to either stationary material or transportable shade at the planter's discretion. When neither stationary nor transportable material is available, the tree may be planted in the open.

Protection Only Requirements (P.1)

Trees shall be planted to protect seedlings from animal trampling, harsh winds and to provide insulation as specified in task order Detailed Information Sheets. Protection material is natural material that acts as an obstacle protecting the tree from animal damage or other injurious agents and may consist of such materials as downed logs, stumps and standing snags. The protection material and effort required by the planter to protect the tree varies depending on the site conditions of each unit. Each tree shall be planted as close to the stump, log or other suitable protection material as possible. It is acceptable to plant on any side of the stationary material when protection-only is required.

P.1.1-Stationary Material Only

Tree shall be planted next to stationary material when it is available. When stationary material is not available, no tree shall be planted.

P.1.2-Stationary Material First, Then Plant in Open

A tree planted shall be planted next to stationary material when it is available. When stationary material is not available, tree may be planted in the open.

Trees not Meeting Specifications

The Contractor is not obligated to plant trees that do not meet the specifications and shall be returned to the Government. Except as noted below, if trees not meeting specifications are planted, they shall be planted satisfactorily.

- In no case shall bareroot trees with root systems less than eight inches long and fewer than three roots (vertical or lateral) be planted.
- Container trees shall have a minimum top height of three inches with the exception of high elevation five-needle pines. Roots shall have a defined main root and still maintain some soil media.
-

Tree Planting Line-Item Descriptions

Regular Spacing Planting (per acre)

Trees shall be planted to achieve the average spacing specified on task order Detailed Information Sheets. It is expected that an irregular distribution will result as planters avoid acceptable established seedlings or trees, rock or shallow soils, and seek planting spots in the best microsites and soil conditions. For individual trees, the spacing may be varied as much as 75 percent in any direction to attain the best plantable spot unless otherwise stated in task order Detailed Information Sheets. The number of trees planted shall not be materially changed by the method of selecting planting spots.

Variable Density Planting (per thousand tree)

Trees will be planted with variable densities based on aspect. Favorable aspects (west to northeast aspects (270 -45 degrees)), shall be planted with higher densities while less favorable aspects (northeast to west aspects (45-270 degrees)) shall be planted with lower densities. Hand compasses (adjusted for declinations listed on task order Detailed Information Sheets) shall be used to determine aspects. Desired aspects and densities to be planted shall be identified on task order Detailed Information Sheets.

Group/Clump Planting (per thousand tree)

This type of planting is designed to provide a more natural arrangement of trees and take advantage of limited resources in harsh planting sites. Trees will be planted in clumps/groups with defined edges with unplanted space left between groups/clumps. Task order Detailed Information Sheets will describe the average number of trees per acre to be planted within the unit as a whole. Density may vary between groups/clumps based on planting spots available within each group/clump. The average number of trees per acre across the unit may not be materially changed. For example, one group/clump may have as few as 75 trees per acre planted while another may have 125 trees per acre planted, but the average number of trees per acre is 100. Individual task orders will provide detail on clump/group locations, the distance between clumps/groups, and the average number of trees to plant per acre.

Per Thousand Tree Planting

This type of planting will consist of planting trees as specified in task order Detailed Information Sheets to take advantage of the best available microsites within units. Planting spot selection takes precedence over spacing requirements.

Interplanting (per thousand tree)

This type of planting consists of planting only a portion of units that have not fully reforested naturally after a disturbance or after prior planting attempts. Task order Detailed Information Sheets will specify the species and densities to be planted intermixed among acceptable established seedlings and acceptable established trees.

Per Hour/Specialized Tree Planting

A flat rate per hour will be paid for each worker and Contractor's on-site representative/Supervisor

Crew size and planting specifications will be described in task orders. Contractor time will start when work commences (i.e., tree bag-up) at access or delivery points, except as otherwise noted in task orders. Examples of this type of work include but are not limited to:

- Planting in timber harvest units or wildfire areas for which the exact acreage is not known.
- Planting areas that require considerable walk-ins which will be compensated for at an hourly rate. This may also require carrying tree boxes or other materials to the planting site itself.
- Planting high value seedlings to ensure trees are planted in the best available microsites.
- Revegetation of campgrounds utilizing a variety of native grass, shrub or tree species with a variety of stock types.
- Planting of Genetic Tests. Planting stock could vary and may be a single stock type or a mix of any stock types of varying ages. Planting will occur on a specified grid and trees will be required to be planted within one foot of pre-designated locations. This may require site preparation such as removal of woody debris and/or clearing of brush.
- Planting in the forest seed orchards or tree improvement areas. Planting stock is typically in one-gallon pots planted on a variety of substrates.

Artificial Shade Installation, Maintenance, or Removal (Tubing)

Shade tubes will require on site construction. Installation of shade (rigid) tubes may be performed in conjunction with the tree planting operation or may be performed as a separate operation but is to be accomplished within 24 hours of planting. Shade tubes shall be installed so as not to damage the tree including, but is not limited to, skinning the bark, breaking the leader, damaging the root, or excessive bud stripping.

Rigid shade protection tubes will be installed using one bamboo stake attached with zip strips to the tubing. Stakes will be forced into the ground to a depth of six inches.

Animal Damage Protection (ADC) Installation, Maintenance, or Removal (netting or tubing)

Flexible or rigid netting shall be installed around trees to protect young seedlings from being browsed by big game. Netting shall be at least one inch and no more than two inches above the terminal leader, and, as required by product, be secured less than one inch from the ground when installed. Netting that is too long shall be folded up at the bottom to achieve desired position. During maintenance, the height of the rigid or flexible netting shall be adjusted and maintained one inch and no more than two inches above the terminal leader to prevent animals from browsing the seedling. Refer to Exhibit 06 for an example diagram of properly installed flexible netting.

Netting shall be installed so as not to damage the tree, which includes, but is not limited to, skinning the bark, breaking the leader, damaging roots or excessive bud stripping. Tree leader must be free to grow, and branches shall be upward and generally in a natural configuration.

Flexible netting shall be applied in a timeframe before big game use in the area. Netting shall be installed or maintained after the growing season, generally the beginning of September and before hunting season in mid-October.

Rigid netting will be installed using one bamboo stake attached with zip strips or woven three times through the tube (see example in Exhibit 06). Stakes will be forced into the ground to a depth of six inches. Installation of rigid netting may be performed in conjunction with the tree planting operation or may be performed as a separate operation to be accomplished within 24 hours of the planting. Rigid netting maintenance will be done on units that had rigid netting installed the previous year.

Performance Requirements Summary

Performance Objective	Required Task	Performance Standard	Method of Surveillance	Positive/Negative Incentive % for NOT meeting the Task
Tree Handling, Care, and Transportation	See <i>Section 4.0 – Tree Handling and Care</i>	Trees are properly handled and cared for. Box temperatures are below 45 degrees F while in the Contractor's possession. When violations occur, Contractor takes corrective action immediately.	COR and GTM will monitor operations and conduct visual inspections.	- The Contractor's care and handling of trees will affect the performance assessment as either a positive or a negative incentive. - If Contractor meets the standard, no deduction. - If Contractor does not meet the standard and/or take corrective action, the Government may assess liquidated or actual damages (see <i>Section 10.0 – Measurement, Remeasurement, and Payment Method - Damages</i>) and provide adverse performance evaluation.
Tree Planting	See <i>section 4.0 – Tree Planting</i>	Planting quality meets or exceeds performance standard of 93% quality.	COR or GTM conduct periodic inspection consisting of plots consistent with Planting Inspection.	- The Contractor's planting quality will be reflected in the performance assessment as either a positive or a negative incentive. - If Contractor achieves 93% planting quality or greater, full unit price as incentive. - If Contractor does not achieve at least 93% quality, deductions will be made as per <i>Section 10.0 – Measurement, Remeasurement, and Payment Method – Method of Payment – Tree Planting Payment Calculation</i> and may result in adverse performance evaluation.

Performance Objective	Required Task	Performance Standard	Method of Surveillance	Positive/Negative Incentive % for NOT meeting the Task
Shade or ADC	See sections 4.0 – <i>Artificial Shade, Installation, Maintenance or Removal (Tubing)</i> and ADC <i>Installation, Maintenance or Removal</i>	Shade or ADC quality meets or exceeds performance standard of 93% quality.	COR or GTM conduct periodic inspection consisting of plots consistent with Shading/ADC Inspection.	- The Contractor's planting quality will be reflected in the performance assessment as either a positive or a negative incentive. - If Contractor achieves 93% quality or greater, full unit price as incentive. - If Contractor does not achieve at least 93% quality, deductions will be made as per <i>Section 10.0 – Measurement, Remeasurement, and Payment Method – Method of Payment - Netting or Tubing Payment Calculation</i> and may result in adverse performance evaluation.
Quality Control	See <i>section 8.0 – Inspection and Acceptance - Contractor Quality Control</i>	Contractor maintains good quality control on all aspects of operation including tree handling, preparation and planting, artificial shade and animal damage control (ADC) treatments. Actions not consistent with contract specifications are corrected as per the Quality Control Plan.	COR and GTM will monitor operations and conduct visual inspections.	- The quality of Quality Control provided by the Contractor will be reflected in the performance assessment as either a positive or a negative incentive. - If the Contractor fails to meet standards, work may be suspended, or the Contractor put in Default for failure to take corrective action immediately and result in adverse performance evaluation.
	See sections 8.0 – <i>Inspection and Acceptance - Contractor Quality Control</i> and <i>Government Quality Assurance Surveillance Plan</i> , and <i>Section 10.0 – Measurement, Remeasurement, and Payment Method – Method of Payment – Work Quality</i>	Contractor's plot inspections are +/-5% of Government's quality. Inspection completed to meet timeframes. Inspection records are complete and accurate.	COR or GTM randomly sample and observe Contractor inspection. Reviews Contractor inspection sheets daily.	- The ability of the Contractor to meet the standard will be reflected in the performance assessment as either a positive or a negative incentive. - When Contractor's plot inspections meet standards, no deductions.

<i>Performance Objective</i>	<i>Required Task</i>	<i>Performance Standard</i>	<i>Method of Surveillance</i>	<i>Positive/Negative Incentive % for NOT meeting the Task</i>
Work Plan and Schedule	See Section 6.0 – Deliverable/Schedule and Section 9.0 – Deliveries or Performance – Work Plan and Schedule.	Work meets schedule as per the accepted work plan.	COR and GTM conduct periodic monitoring.	• The Contractor's ability to maintain the work schedule and complete work on schedule will be reflected in the performance assessment as either positive or negative incentive. • Work not completed on schedule is subject to Default and may result in assessment of actual costs.

6.0 Deliverables/Schedule

The following table provides a summary of deliverables with respective due dates that need to be submitted by the Contractor.

Deliverable	Objective/Contract Section Reference	Due
Quality Control Plan (QCP)	Describes how Contractor's quality will be monitored to assure appropriate tree handling and care, that performance standards are met, the supervision of crew and work, and personnel that will be performing the quality control. Reference <i>Section 8.0 – Inspection and Acceptance - Contractor Quality Control – Quality Control Plan</i> .	With technical proposal
Safety Plan	Describes measures the Contractor takes to keep employees safe under a variety of normal and unexpected situations. Reference <i>Section 9.0 – Deliveries or Performance – Safety Plan</i>	With technical proposal and updated prior to commencing work on any task order.
Work Plan and Schedule	The Contractor shall prepare a work plan and schedule that demonstrates how the work will be completed within the period of performance identified. Reference <i>Section 9.0 – Deliveries or Performance – Work plan and Schedule</i> .	With technical proposal and updated prior to commencing work on any task order.
Worksite Plan	The Contractor shall prepare a worksite plan describing how the worksite will be maintained during operations. Reference <i>Contract Section XX</i> .	With technical proposal and updated prior to commencing work on any task order.
Contractor Quality Control Inspections	Completed and legible quality control inspection forms for tree planting and artificial shade/ADC treatments. Reference <i>Section 8.0 – Inspection and Acceptance - Contractor Quality Control – Quality Control Plan</i> .	At the end of each day's work for subitems completed that day

7.0 Contractor's Key Personnel

The Contractor shall comply with the following requirements, unless otherwise approved by the CO, or mutually agreed to in writing by the CO and the Contractor. The crew configuration information identified here does not dictate required production rates.

Supervisor

At all times during performance of this contract and until task order work is completed and accepted, the Contractor shall directly superintend the work or assign and have on the worksite a competent superintendent who is satisfactory to the CO and has authority to act for the Contractor.

The Contractor shall designate at least one supervisor per task order. This supervisor shall not perform tree dipping, tree planting, animal damage control, or shade work.

The Contractor or his on-site representative shall provide a supervisor for each planting crew of up to 15 planters, unless otherwise authorized by the CO. The supervisor must be bilingual and able to read and speak English fluently.

Inspectors

Contractor Inspectors shall not perform any tree dipping, tree planting, animal damage control, or artificial shade work on any calendar day that they are assigned to Inspector duties.

Language

The Contractor's on-site representatives, inspectors, and all seedling runners/carriers shall be fully conversant in the English language. The representative, inspectors, and runners/carriers shall be able to fully comprehend English and interpret English instructions fully and completely to the native tongue of his crew.

Organization of Crews

Planting shall be performed in an organized systematic manner. Crew members shall not be scattered within or between units except as necessitated by on-the-ground conditions and only when authorized by the COR.

8.0 Inspection and Acceptance

Contractor Quality Control

The Contractor shall provide quality control for all planting, dipping, animal damage control, and artificial shade operations, ensuring that seedlings are handled in accordance with *Section 4.0 - Tree Handling and Care* specifications. The Contractor is also responsible for the quality of planting on site and is expected to conduct their own prompt inspections to ensure proper planting procedures are being followed. Government inspections, as described below, are for payment determination only, and are not intended to be used as a quality control by the Contractor. Any delay in Government inspections will not cause the Government to be responsible for the Contractor's planting quality.

Quality Control Plan

The Contractor shall provide a general Quality Control Plan (QCP) with the contract technical proposal. The Contractor shall supplement the QCP within 30 days prior to starting work, based on items awarded to the Contractor and Government comments. The general QCP must demonstrate how the Contractor will assure quality meets the requirements and specifications of the contract. The QCP shall describe

how quality will be monitored to assure appropriate tree handling and care, the performance standards are met, the supervision of crew and work, and personnel that will be performing the quality control. Quality control shall include monitoring tree care and other associated tasks, quality while planting and artificial shade/ADC treatments are in progress, and an unbiased sample of planting, shading and/or ADC treatments for determining actual quality percentage.

Per Planter/Specialized Tree Planting (Per Person per Hour Unit of Measure)

The QCP shall demonstrate an understanding that hourly work items will be directed by the Government and paid out at the per person per hour rate and that any hourly work not meeting contract specifications/standards that require rework will be monitored and documented by the Government.

Any rework will not be compensated for under this bid item.

Inspections

Inspection shall be maintained concurrent with work. Contractor inspection results shall be available to the COR at the completion of each day's work for subitems completed that day. Completed Planting Inspection Sheets (see Exhibit 07), Artificial Shade/ADC Inspection Sheets (see Exhibit 11), and sketch maps remain a part of the contract file.

The Contractor shall submit legible inspection sheets and work maps. Illegible and incomplete inspection sheets will be returned to the Contractor for correction and resubmission before Government verification inspections are performed.

By submitting completed inspection sheets to the Government, the Contractor is certifying that the subunit/unit is:

- Satisfactorily completed.
- Tree handling was completed in accordance with the contract provisions stated elsewhere in the contract.
- Planting or Artificial Shade/ADC inspection reflect the work accomplished.

Plots shall be marked and identifiable on the ground. Identify inspection plot centers on the ground with flagged markers. Mark flagging in permanent marker with unit name and number, plot number, date of inspection and inspectors' initials. An additional piece of flagging shall be hung at least 4 feet above ground adjacent to marked plot center.

The Contractor shall inspect artificial shade/ADC installation (except for flexible netting), where required, concurrently when inspecting tree planting.

Tree Planting

Per Acre Unit of Measure

The Contractor shall maintain a plot system that provides an unbiased sample of planting quality and which represents the entire area planted. The plot system shall be consistent with the planting inspection method described in Exhibit 09 – Contract Inspection Method using the Planting Inspection Sheet (Exhibit 07) and associated instructions (Exhibit 08). The Contractor shall sample at least one percent of the planted area, with a minimum of five plots per subitem. A 1/20th, 1/50th, or 1/100th acre plot size shall be used during inspection, depending on tree planting density and slope of sub-units. The Government may observe or re-examine the Contractor's plots at any time. Each subitem will be assessed separately and results will not be averaged with those of other subitems.

Plot sampling shall be done on an unbiased, uniform, right angle grid located throughout the planting area to obtain a representative sample of the work.

Calculation of planting quality will be based on the following formula (refer to the columns of the Planting Inspection Sheet in Exhibit 07). All computations will be carried out to at least thousandths (.000) and the final percentage of planting quality will be rounded to the nearest whole percent:

$$\frac{\text{Number of Satisfactory Trees Above Ground (Column 9)}}{\text{Number of Plantable Spots (Column 5)}} \times \frac{\text{Number of Satisfactory Trees Below Ground (Column 11)}}{\text{Number of Dug Trees (Column 10)}} \times 100$$

Plot inspection shall include a tally of trees that are planted in excess of the maximum allowable trees on each plot and noted in Column 8 of the planting inspection sheet. These excess trees are considered wasted trees. The sum of wasted trees on all plots on the subitem will be used to determine the damage assessment.

Per Thousand Tree Unit of Measure

The Contractor shall maintain a plot system that provides a sample of planting quality representative in portions of subunits where trees were planted (this can be a biased sample). Plots should be placed as uniformly as possible in areas where trees were planted, with a grid system recommended but not required. The plot system shall be consistent with the planting inspection method (Exhibit 09), using the Planting Inspection Sheet (Exhibit 07). The Contractor shall sample at least one percent of the planted area, with a minimum of five plots per subunit. A 1/20th, 1/50th, or 1/100-acre plot size shall be used during inspections depending on planting density and slope of subunits. The Government may observe or re-examine the Contractor's plots at any time. Each subitem will be assessed separately and results will not be averaged with those of other subitems. Reference *Section 8.0 – Inspection and Acceptance - Contractor Quality Control – Inspections* and the *- Per Acre Unit of Measure* sections for plot identification, inspection sheet/sketch map submittals, and calculation of planting quality percentage.

Variable Density Planting

The average spacing required by aspect as listed on task order Detailed Information Sheets, in combination with plot size, will be used to determine the desired number of plantable spots (Column 3 on the Planting Inspection Sheet) by aspect on any given plot (this may vary by plot). Wasted trees will be determined as described in Exhibit 08. The sum of wasted trees on all plots on the subitem will be used to determine the damage assessment.

Group/Clump Planting, Per Thousand Tree

The desired number of plantable spots (Column 3 on the planting inspection sheet) will be determined at each plot based on the number of available visible microsites within a plot. If appropriate microsites are available and if no shade or protection violations (code 'X' on Exhibit 07) are noted on a plot, more trees than the maximum trees per plot can be planted on each plot and will not be considered wasted trees. If shade or protection violations are noted on any plot, these will be considered wasted trees. The sum of wasted trees on all plots on the subitem will be used to determine the damage assessment.

Interplanting

The desired number of plantable spots (Column 3 on the planting inspection sheet) will be determined at each plot based on the number of existing acceptable seedlings and trees within each plot. Wasted

trees are those that are planted closer than minimum distances away from existing acceptable seedlings and trees within each plot. The sum of wasted trees on all plots on the subitem will be used to determine the damage assessment.

Shading/Animal Damage Control

The Contractor shall sample at least one percent of the planted area, with a minimum of five plots per subunit. A 1/20th, 1/50th, or 1/100-acre plot size shall be used during inspections depending on planting density and slope of subunits and record results on the Artificial Shade/ADC Inspection Sheet (Exhibit 11). Each sub-item will be assessed separately, and results will not be averaged with those of other subitems.

The Government may observe or re-examine the Contractor's plots at any time. Each subitem will be assessed separately and results will not be averaged with those of other subitems. Calculation for percent quality for artificial shade and/or ADC treatments will be based on the following formula:

$$\frac{\text{Satisfactory Trees}}{\text{Total Trees Requiring Treatment}} \times 100$$

The inspection shall include a tally of wasted materials due to: 1) excess or unspecified material used or treated within the inspection plot; and 2) materials dropped or damaged by Contractor. Wasted material shall be tallied by plot. The Government will use the sum of all plots on the subitem to determine the damages assessment.

When Included with Tree Planting Inspection

The same plots and identification methods used for inspecting tree planting shall be used for artificial shade/ADC installation.

When Not Included with Tree Planting Inspection

When Tree Planting Using per Acre Unit of Measure

The Contractor shall maintain a plot system that provides an unbiased, uniform, right angle grid-based sample of artificial shade/ADC installation quality which represents the entire area treated.

When Tree Planting Using per Thousand Tree Unit of Measure

The Contractor shall maintain a plot system that provides a sample of artificial shade/ADC installation quality representative in portions of subunits where trees were planted (this can be a biased sample). Plots should be placed uniformly in areas where trees were planted, with a grid system recommended in these areas but not required. For the average installation or maintenance trees per acre (TPA) on the header of the inspection sheet, the Contractor will use the average number of trees per acre that were planned for the respective portions of the subunit as specified on task order Detailed Information Sheets.

Government Quality Assurance Surveillance Plan (QASP)

The Government will assure the Contractor is providing the quality promised and that work meets the performance standards. Surveillance will consist of visual observations of the entire operation and will include dipping (when applicable), tree handling and care, planting operations, and artificial shade/ADC treatment operations. A series of inspection plots to evaluate the quality of planting and artificial shade/ADC treatments (when applicable) shall be performed. The Government will use the Planting Inspection Sheet (Exhibit 07) or Artificial Shade/ADC Inspection Sheet (Exhibit 11) and follow the

procedures outlined for Contractor inspection. Plot densities and locations, however, may differ from that done by the Contractor. Each subitem/unit will be verified separately and inspection results on one subitem/unit will not be averaged with those of other subitem/units.

Trees that are mishandled, dropped, or stashed will be declared wasted trees.

Daily, the Government will document the number of trees that the Contractor requests, is issued, and returns at the end of the workday. A receipt for trees (see Exhibit 12, Tree Tacking Sheet) issued and returned must be signed by, and agreed upon, by both the COR/inspector and the Contractor/Contractor's Representative prior to end of planting on each sub item.

Tree Quantity for Per Thousand Tree Unit of Measure (UOM)

The COR may determine tree quantity by taking a series of "count" plots, where trees planted and wasted (excess trees) are tallied using the Planting Inspection Sheet. The COR may also use a count of the boxes of trees issued to the Contractor and planted on a particular day. A minimum two percent sample will generally be taken. The Government's plots taken for purposes of quality assurance may also be used for the determination of quantity of trees planted. The Government shall investigate large discrepancies between plot calculations and trees issued.

Per Planter/Specialized Tree Planting (Per Person Per Hour Unit of Measure)

Work under this item will be directed by the Government. Quality control will be visually inspected. Methods similar to those described in *Section 8 – Inspections and Acceptance* may also be used at the Government's discretion. If quality is unacceptable, the Government will inform the contractor for immediate correction at the job site.

9.0 Deliveries or Performance

Contract Period of Performance

The task order period of this bid item is from the date of contract award through the last period identified by the Government.

Contract Year	Start of Contract Option Date	Last Date a Task Order May Be Executed
2026	Award date of contract	June 15, 2026

Work Plan and Schedule

The Contractor shall prepare a work plan and schedule that demonstrates how the work will be completed within the period of performance identified. Work plans and schedules will be updated upon receipt of each task order. It is the Contractor's responsibility to promptly submit the Work Plan and Schedule. No adjustments in contract time or the start work date for the task order will be made solely due to the failure of the Contractor to submit a Work Plan and Schedule.

The Contractor shall advise the CO of any periods that the Contractor will not be working that is not specified in the accepted Work Plan.

The Contractor may update the Work Plan when work is in progress and the CO may accept the updated work plan.

If more than one area is available for planting, the CO may prescribe the planting sequence.

Safety Plan

The Contractor shall provide a copy of their safety plan, which will be kept in the contract folder for OSHA's review. The Safety Plan will be updated for each awarded task order. and submitted upon receipt of task orders.

The Safety Plan should be available to all employees and subcontractors working on the contract.

The Safety Plan should cover at a minimum:

- Safety briefings covering the work area, equipment being used, and any special considerations for the work under. Recognize that some planting units may be in areas of snags and/or fire burned trees. The Contractor shall identify the specific conditions that the Contractor considers a hazard and shall include working in areas of dead trees and falling debris.
- Personal Protective Equipment (PPE) required for the work.
- Identify appropriate OSHA requirements for the type of work.
- Weather considerations for the time of year.
- Communication issues and locations of the nearest medical facilities.
- Name and point of contact information of supervisor responsible for enforcing Safety Plan.
- Name, position title and point of contact information of company executive responsible for approving the Safety Plan.
- Specific contingency plans for emergencies such as medical, fire, hazard material spills and any other contract assessed hazard prevention and abatement requirement needs that apply to this project.
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Task Order Period of Performance

Task Order Period of Performance

Period of performance for individual task orders will be identified in each task order. Periods of performance time for multiple task orders issued may overlap and/or run concurrently.

Estimated Start Work Date

The estimated start work date for each task order will be identified when task orders **are solicited** by the CO.

Use and Occupancy Restriction Orders

Bear Safety

The following National Forests have Use and Occupancy Orders for bear safety; these orders shall be adhered to while working on these agencies:

- [Idaho Panhandle National Forests:](#)
- [North Zone:Bonners Ferry, and Sandpoint Ranger Districts.](#)

Violation by the Contractor, his agents and employees, of the Endangered Species Act of 1973 on lands covered by and used in conjunction with this contract will be cause for termination of this contract.

The Contracting Officer or his representatives may order an immediate temporary suspension of all activities within work areas if needed to resolve potential or existing grizzly/human conflicts. The Contractor shall immediately comply with this direction. It should be noted that the Government shall

not be liable for any consequences from any such suspensions to contracts that are less than ten days in length.

The Contractor assumes full responsibility and shall hold the Government harmless from any and all claims by the Contractor, subcontractor, or third parties for any damage to life or properties arising from the activities authorized by this contract and encounters with grizzly bears from suspensions of activities authorized by this contract. Intentional or negligent acts by the Contractor or subcontractors, employees, or third parties that result in injury or death of a grizzly bear will be cause for suspension or termination of this contract in whole or part.

Specific to the listed ranger districts or portions of the ranger districts of the Idaho Panhandle National Forests:

- Garbage, unused foods, petroleum products, antifreeze and other bear attractants shall be removed from these agency's lands daily or stored in acceptable bear-resistant containers on these agency's lands while not in use from March 1st to December 1st. Refer to [Bear-Resistant Products - IGBC Online](#) for acceptable sources of bear resistant containers.
- No personnel should intentionally approach closer than 100 yards to a grizzly bear.
- Feeding bears for any purposes is prohibited
- Any bear activity within or adjacent to camping or work areas will be reported to the Contracting Officer or his representatives within 12 hours.
- Hard sided camping only may be required in certain areas within these agency's boundaries. Task orders will list this requirement if applicable.

Notice to Proceed

The Government will issue a Notice to Proceed as soon as weather and ground conditions are favorable for planting and survival of the trees on one or more of the areas. If more than one area is available, the Contractor may specify the unit sequence with agreement by the CO.

Contract time will start 1) on the date specified on the Notice to Proceed, if the notice is hand delivered, faxed, or e-mailed; 2) on the third day after the post office has issued a Notice of Certified Mail to the Contractor; or 3) when planting commences, whichever is soonest. Failure of the Contractor to pick up certified mail will not be considered an excusable delay.

The CO may, at the CO discretion, agree to a Contractor-proposed specific start work date if the CO determines that the date proposed by the Contractor is acceptable to the Government.

Minimum Rate of Progress

Tree survival and growth is dependent upon proper soil and weather conditions at the time of planting and these conditions only prevail for a limited length of time. It is therefore imperative that work be started promptly after conditions for planting become suitable.

The Contractor shall maintain progress at a rate, which will assure completion within the task order period of performance. The minimum acceptable rate of progress will be calculated by dividing the total units of work for each item by performance time specified.

In the event the Contractor is delayed and falls more than two calendar days behind the progress rate required above, the Contractor's right to proceed is subject to immediate termination for default.

Unfavorable Conditions

Whenever it is determined that environmental or physical conditions become unsuitable for tree survival, or when extreme weather conditions create the potential for blowdown, the Contractor shall move to another area identified by the Contractor or the CO, or take other mitigating actions directed by the CO. When no other area is available or mitigating actions to be taken, the Government shall suspend work and stop the contract time. When conditions are again favorable, the CO will issue a verbal resume work order and confirm that order in writing. The count of contract time will resume on the day indicated by the CO. It is the Contractor's responsibility to keep the CO informed of how the Contractor or Contractor's representative can be reached. If the Contractor cannot be reached, a written Resume Work Order will be mailed and the count of contract time will resume on the date specified in that work order.

Work Completed After the Performance Period

Whenever the Contractor fails to complete work within the task order period of performance, and works beyond that period, the Contractor will be assessed actual damages. Items to be included in the actual damage includes, but is not limited to, wages of the COR and Government Inspector(s) for quality assurance after the performance period, costs of vehicles, meals, and lodging that are incurred while administering the contract after the performance period.

10.0 Measurement, Remeasurement, and Payment Method

Measurement

All linear and area measurements under this contract are measured on a horizontal plane.

Per Acre Unit of Measure

The Contractor will be paid based on the acres indicated in task orders. The method of acreage determination will be specified in task orders (i.e., GIS, GPS, differentially corrected GPS). When acreages provided on task orders are estimates, it will be labelled as such, and the COR will provide the correct acres to the Contractor prior to invoicing.

Per Thousand Tree Unit of Measure

The actual number of trees planted will be computed based on the Government's plots conducted for this purpose using the following formula. Results will be rounded to the nearest tenth (i.e., 4.2 thousand trees). Note that the reference to 'Columns' refers to those columns on the Planting Inspection Sheet (Exhibit 07).

$$\frac{\text{Total Number of Trees Planted (Column 7)}}{\text{Number of Plots}} - \frac{\text{Total Wasted Trees (Column 8)}}{\text{Number of Plots}} \times \frac{\text{Reciprocal of Plot Size}}{\text{Plot Size}} \times \text{Acres} = \frac{\text{Number of Trees}}{1000}$$

When the calculated number of trees exceeds the number of trees issued, the payment will be limited to the number of trees issued, less any wasted trees.

Walk-Ins

Distances will be measured horizontally from the access point to the unit boundary along the route designated by the Government or the route taken by the Contractor, whichever is the shorter of the two, and will be rounded to the nearest hundredth of a mile (i.e., 1.55 miles).

Remeasurement

The Contractor may at any time after award request re-measurement of any subitem, except for per-person per hour units of measure. If a Contractor requests remeasurement, the same methods will be used to determine acreage as the original method. For example, a planting unit was traversed with differentially corrected GPS, then the remeasurement must utilize differentially corrected GPS. The request must be in writing within ten calendar days after completion of a subitem. The Government's re-measurement of the subitem will be made within the established boundaries.

1. Payment will be based on the re-measurement of the acreage or quantity count except when:
 - Per Acre Unit of Measure: No payment adjustment shall be made for areas less than one acre.
 - Per Thousand Unit of Measure: No payment adjustment shall be made for quantities of 500 or less.
2. The Contractor shall pay for the cost of remeasurement if the remeasurement results in:
 - Per Acre Unit of Measure: an increase of less than one acre on units less than or equal to 20 acres; or an increase of less than five percent on units larger than 20 acres.
 - Per Thousand Unit of Measure: An increase of less than five percent.

Method of Payment

Work Quality

For each completed subitem, the Contractor's quality control inspection percentage for tree planting, artificial shade, or animal damage control will be compared to the Government's assessment results. If the results differ by five percent or less, and the inspection has been conducted to the standards specified, the Contractor's inspection will be considered acceptable, and that quality percentage will be used as the basis for payment. When the results differ by more than five percent, the Government's inspection percentage will be used as the basis for payment.

If the Government's results are objectionable to the Contractor, the Contractor may request a re-examination in writing within five calendar days after receipt of the Government's quality assurance inspection results. The request shall include documentation that supports that the probability that an error exists. The Government's re-examination will consist of a two percent sample utilizing the same inspection method as the original examination, but new plots will be selected with different plot centers than the plots previously inspected by the Government. If the results of the re-examination are within five percent of the Contractor's submitted quality control results, the Contractor's results will be considered acceptable and used for payment. If results of the re-examination exceed five percent of the Contractor's submitted quality control results, the Government's re-examination results will be used for payment and the Contractor will be assessed the costs incurred by the Government that are directly related to the performance of such service. The Government is not required to perform a re-inspection.

No reinspection will occur for subitems utilizing the per person per hour unit of measure. Any rework will be completed at the Contractor's expense.

Tree Planting Payment Calculation

Payment, less damages, is based on work quality percentages described below.

Quality Meets or Exceeds 93 Percent

Whenever the quality of planting meets or exceeds 93 percent, based on the Contractor's quality control inspection (plots) and verified by the Government assessment, the quality of planting will be acceptable, and payment will be made at the full unit price for tree planting. Whenever the Government's quality of planting is below 93 but within five (5) percent of the Contractor's quality control inspection (plots) full payment shall be made.

Payment Example: With a unit price of \$100 and a unit size of 60 acres, the bid price would be \$6,000. If the Contractor's planting quality was 95 percent, payment would be \$6,000.

Quality less Than 93 percent and at least 80 Percent

Whenever the quality of planting is below 93 percent but is 80 percent or over, based on the Contractor's quality control inspection (plots), and verified by the Government monitoring/assessment the work will be considered unacceptable, and the value of the services is reduced. Therefore, payment will be made after a one percent deduction in the unit price for each one percent the planting quality is below 100 percent. This payment will only be made after the Government determines that no further work shall be required, nor allowed.

Payment Example: Payment Example: With a unit price of \$100 per acre and a unit size of 60 acres, the bid price would be \$6,000. If the planting quality were 85 percent, payment would be $\$100 \times 85\% \times 60 \text{ acres} = \$5,100.00$.

Quality Less than 80 Percent

Whenever the quality of planting is below 80 percent, based on the Contractor's quality control inspection (plots) and verified by the Government monitoring/assessment, the quality of planting will be considered unacceptable. No payment will be made where quality of work is below 80 percent.

Netting or Tubing Payment Calculation

Payment, less damages, will be based on the methods described below.

Quality Meets or Exceeds 93 Percent

Whenever the quality of artificial shade or ADC treatments meets or exceeds 93 percent, based on the Contractor's quality control inspection (plots) and verified by the Government assessment, the quality of planting will be acceptable, and payment will be made at the full unit price for artificial shade or ADC.

Payment Example: With a unit price of \$30 per thousand and 60.0 thousand trees tubed or protected, the bid price would be \$1,800. If the Contractor's quality were 95 percent, payment would be \$1,800.

Quality less Than 93 percent and at least 80 Percent

Whenever the quality of artificial shade or ADC treatments is below 93 percent but is 80 percent or over, based on the Contractor's quality control inspection (plots) and verified by the Government monitoring/assessment, the work is unacceptable, and the value of the services is reduced. Therefore, payment will be made after a one percent deduction in the unit price for each one percent the shading or ADC quality is below 100 percent. This payment will only be made after the Government determines that no further work shall be required, nor allowed.

Payment Example: With a unit price of \$300 per thousand and 60 thousand trees to be treated, the bid price would be \$1,800. If the quality were 85 percent, payment would be $\$30 \times 85\% \times 60 \text{ thousand trees} = \$1,530$.

Quality Less than 80 Percent

Whenever the quality of artificial shade/ADC treatment is below 80 percent, based on the Contractor's quality control inspection (plots) and verified by the Government monitoring/assessment, the quality of artificial shade/ADC treatments will be considered unacceptable. No payment will be made where quality of work is below 80 percent.

Per Person Per Hour Unit of Measure

A receipt for the number of hours worked and the number of workers on site must be signed by, and agreed upon, by both the COR/inspector and the Contractor/Contractor's Representative prior to the end of work on this subitem.

Payment example: With a unit price of \$15.00 per hour per person, for 20 hours of work for a crew of 15 people (including the Supervisor), payment would be calculated as $\$15.00 \times 20 \times 15 = \$4,500$.

Unacceptable Work

When the Contractor delivers work determined to be unacceptable, the Government may:

- accept the work at a reduced price;
- require the area to be reworked at no additional expense to the Government; or
- reject the work in its entirety. If the work quality is unacceptable, the Contractor's right to proceed is subject to immediate termination for default.

The Government will permit the Contractor to rework an area only if planting stock or other material is available and/or the existing deficiencies can be corrected. Rework will not be allowed with the per person per hour unit of measure.

When planting, artificial shade, or ADC treatment quality is considered unacceptable for areas one acre or larger and the quality of work can be corrected, the Government may require the area to be reworked. If the Government does not require the area to be reworked or the quality remains below 80 percent, the area may be separated from the sub item and payment made in accordance with *Section 10.0 - Method of Payment* as applicable.

Any Government quality assurance assessment needed due to the Contractor reworking an area, either at the Contractor's request or because of unacceptable work, will be at the Contractor's expense. The Contractor may be assessed actual damages which include but are not limited to: wages of the COR and Government inspector(s) incurred due to the rework; costs of vehicles; and, meals and lodging that are incurred during contract administration of the rework.

Damages

Liquidated Damages

It is understood and agreed that the loss to the Government by lowered planting or shading/ADC treatment quality below 93 percent may likely result in replanting or re-treatment in the future. If the

Contractor fails to plant or treat at an acceptable level of quality (93 percent), the Contractor shall, in place of actual damages, pay to the Government as fixed and agreed liquidated damages for each unit of measure treated at less than 93 percent quality, as described below:

- Less than 80 percent quality = Contractor's bid price per unit
- Quality between 80 and 93 percent = Actual quality % below 100

Alternatively, if quality of planting or treatment quality falls below 80 percent and is not corrected, the Government may terminate this contract in whole or in part under the Termination for Cause clause in this contract and, in that event, the Contractor will be liable for fixed and agreed liquidated damages on all subitems planted where the planting or treatment quality is less than acceptable. In addition to liquidated damages charged for subunits planted or treated where the quality is less than acceptable, the Contractor may be assessed actual damages for the remaining subunits that have not been planted or treated on the contract but that have been terminated.

The Contractor shall not be charged for liquidated damages when the unsatisfactory quality of planting arises out of causes beyond the control and without the fault or negligence of the Contractor as defined in the Termination for Default clause in this contract.

Actual Damages

It is mutually agreed that if the Contractor wastes seedlings or materials at a rate that will not allow the planting or treatment of an entire subitem, the unplanted or untreated areas may be deleted from the contract through a partial default termination. In this event, the Contractor shall be liable for damages suffered by the Government to plant or treat the defaulted subitems whenever additional seedlings or materials become available. In addition, the estimated cost of damages may be withheld from the Contractor's earnings.

Excess Trees

Trees that are tallied on the Plot Inspection Sheet (Exhibit 07) that exceed the maximum allowable trees to be planted (Column 8) are considered wasted trees. If the Contractor's inspection has a planting quality within five percent of the Government's inspection and has a wasted tree total that is within 50% of the Government's total, the Contractor's inspection shall be used to calculate wasted trees. If the Contractor's inspection has a wasted tree total that is not within 50% of the Government's, the Government's inspection shall be used to calculate wasted trees. The total of all wasted trees on the payment unit (tallied by plot), will be used to determine the wasted tree charge for excess trees planted on the sub item based on this calculation:

$$\frac{\text{Sum of Wasted Trees on Plots (Column 8)}}{\text{Number of Plots Taken}} \times \text{Reciprocal of Plot Size} \times \text{Acres in Subitem} \times \text{Charge Rate Per Tree} = \text{Wasted Tree Charge}$$

The CO may waive the wasted tree charge for excess trees planted if it is minor in nature. In no case will a waiver be given if the total "Planted Trees" (Column 7) on all inspection plots for the subitem exceeds the "Maximum Number Allowable Trees" (Column 6) for the sub item.

Mishandled Trees

If the Contractor mishandles trees in such a way that it results in actual or potential damage to the tree(s), the trees(s) will be considered wasted. This includes trees that are stashed, dropped, or damaged by the Contractor while the trees are in the Contractor's possession which may occur at the dipping site, planting site, or in contractor transport. Charge for individual trees that are mishandled is as follows:

$$\text{Number of Wasted Trees} \quad \times \quad \text{Charge Rate Per Tree} \quad = \quad \text{Wasted Tree Charge}$$

When wasted trees are found and the damage is considered minor in nature, the CO may waive the wasted tree charge.

Excess Artificial Shade or ADC Material

Material that is used to protect trees unnecessarily on the unit will be considered wasted material and noted on the plot inspection. The total of wasted material is the sum of all plots for each subitem based on the inspection accepted for payment. The charge for wasted material will be calculated as follows:

$$\frac{\text{No. of Wasted Material on Plots}}{\text{No. of Plots Taken}} \quad \times \quad \text{Reciprocal of Plot Size} \quad \times \quad \text{Acres in Subitem} \quad \times \quad \text{Charge Rate Per Item} \quad = \quad \text{Wasted Material Charge}$$

Mishandled Artificial Shade or ADC Material

Wasted material charge will be assessed when material is stashed, dropped, damaged, mishandled by the Contractor in such a way that it becomes unusable. Charge for wasted material:

$$\text{Number of Wasted Items} \quad \times \quad \text{Charge Rate Per Item} \quad = \quad \text{Wasted Material Charge}$$

When wasted materials are found, and the damage is considered minor in nature, the CO may waive the wasted material charge.

Charge Rates

The charge for wasted trees or materials and other Government-furnished property that is not returned, or returned in a non-usable condition, is listed in the following table:

Item	Charge Rate
Bareroot Seedlings	
• 1-0	\$0.45 each
• 2-0	\$0.48 each
Container Seedlings	
• 160	\$0.49 each
• 112	\$0.575 each
• 91	\$0.653 each

CONTRACT DOCUMENTS, EXHIBITS OR ATTACHMENTS

Exhibit Number	Description
Exhibit 01 (PG 38)	Example 2 Detailed Information Sheet
Exhibit 03-04 (PG 39-40)	Planting bareroot with hoedad
Exhibit 05 (PG 41-42)	Satisfactory and Unsatisfactory Plantings
Exhibit 06-07 (PG 43-44)	Utilization of Material to Provide Shade
Exhibit 09 (PG 45)	Planting Inspection Sheet
Exhibit 10 (PG 46)	Planting Inspection Instructions
Exhibit 11 (PG 47-51)	Planting Inspection Method
Exhibit 12 (PG52-54)	Sample Planting Inspection and Explanatory Notes
Exhibit 13 (PG55-56)	Spanish Planting Inspection sheets and instructions.
Exhibit 14 (57-58)	Tree Tracking Sheet

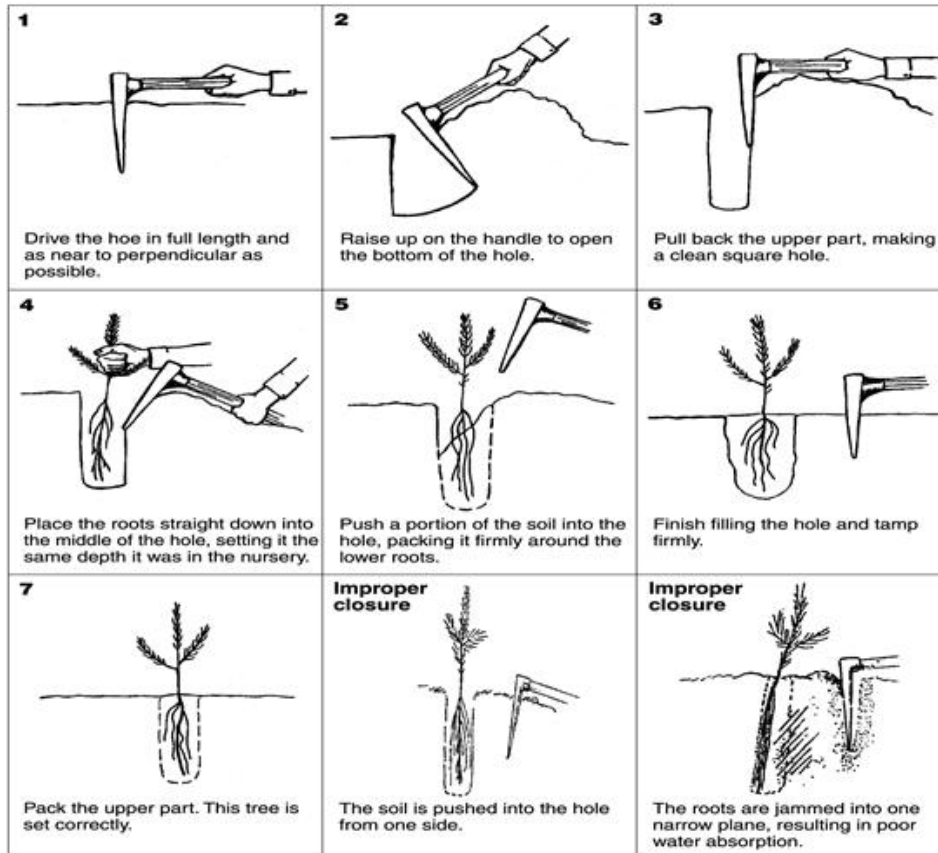
Example 2

[illegible]

Exhibit 03 – Planting Procedures

Hoe planting Procedures

Procedure When Planting With a Hoe



Hand Site Preparation

(centerhole / sidehole examples)

Slope scalp with debris escape route

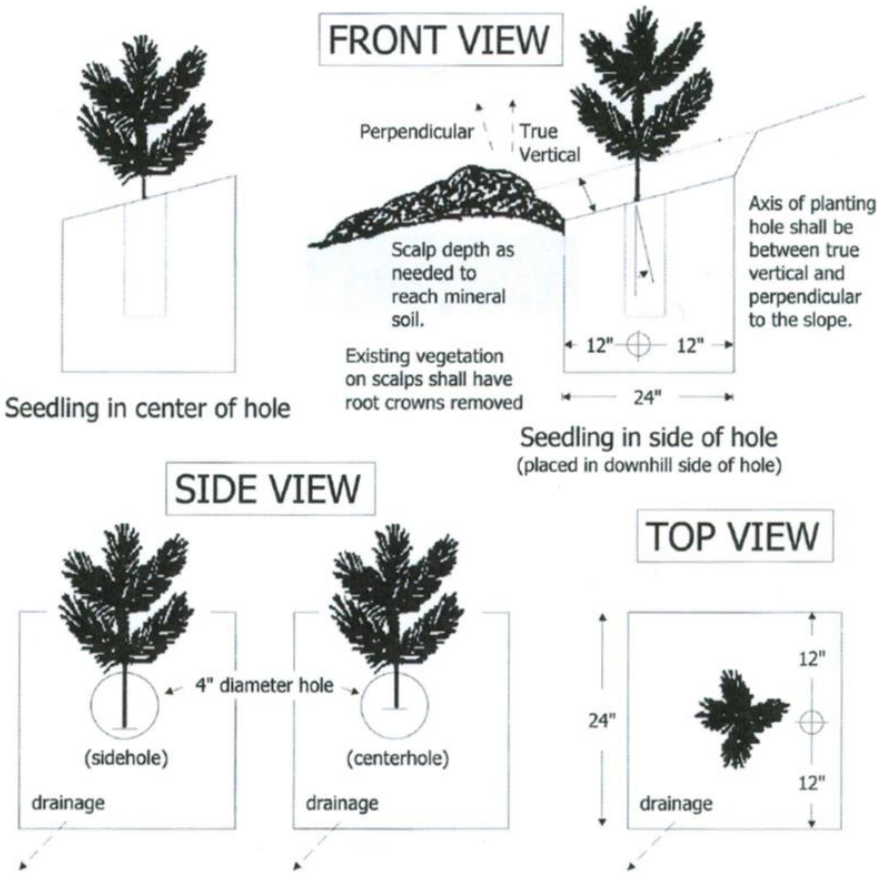
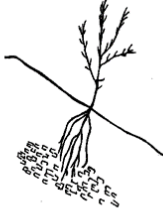
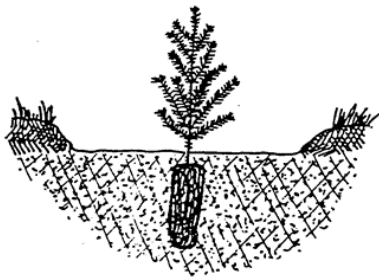


Exhibit 05 – Satisfactory and Unsatisfactory Plantings

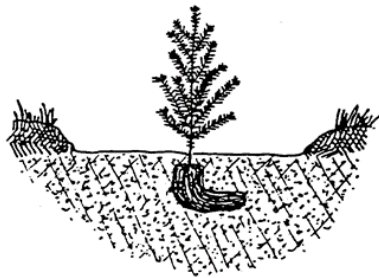
Satisfactory and Unsatisfactory Plantings

SATISFACTORY 	SATISFACTORY 	Unsatisfactory  Too deep. Needles buried.
Unsatisfactory  Improper orientation. Not planted into the slope or near vertical.	Unsatisfactory  "L" roots. Shallow hole.	Unsatisfactory  "J" roots. Shallow hole. Roots often exposed.
Unsatisfactory  Jammed roots. Hole too narrow and shallow.	Unsatisfactory  Hole too shallow. Roots exposed.	Unsatisfactory  Air pocket because of improper tamping.
Unsatisfactory  Planted in rotten wood. Roots not in mineral soil.	Unsatisfactory  "U"- or "J"-shaped tap root.	Unsatisfactory  Compacted roots. Hole too narrow.

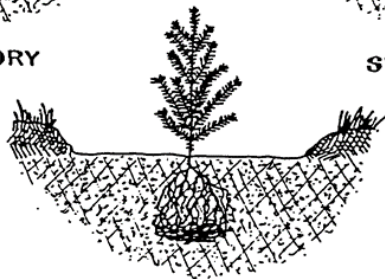
Container Seedling Placement



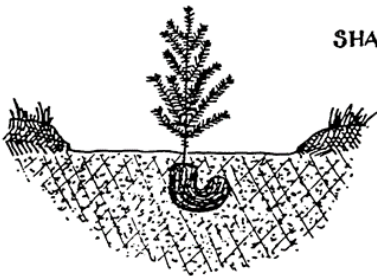
SATISFACTORY



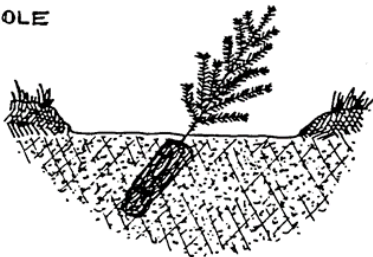
SWEPT ROOT



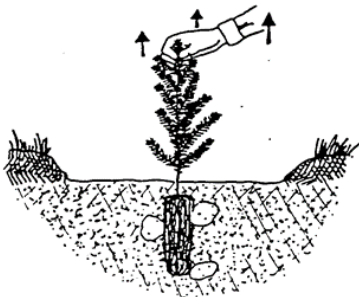
SHALLOW HOLE



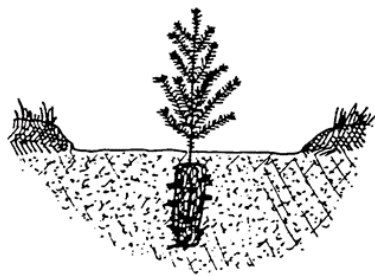
"J" ROOT



LEANER



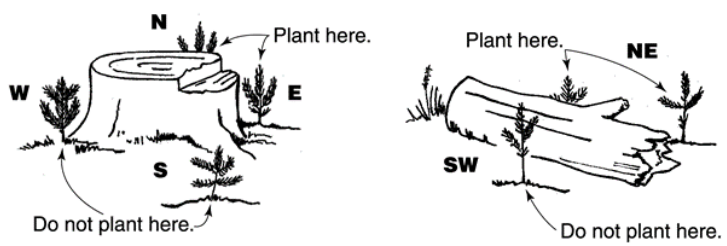
LOOSE/AIR POCKETS



DEBRIS

Utilization of Material to Provide Shade

Using existing shade



Transporting shade

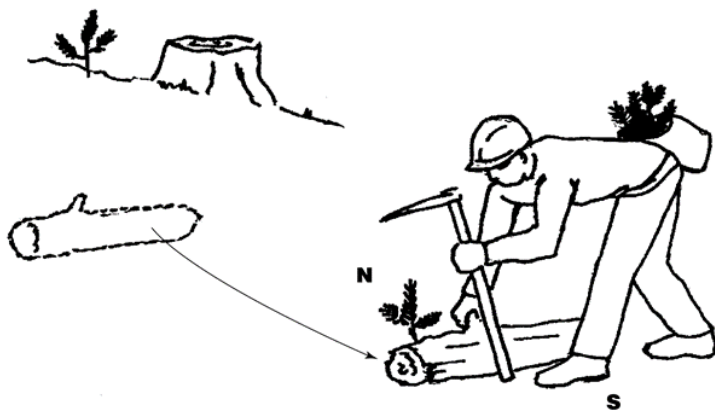


Exhibit 07 – Shade Requirements – Microsite Only Example

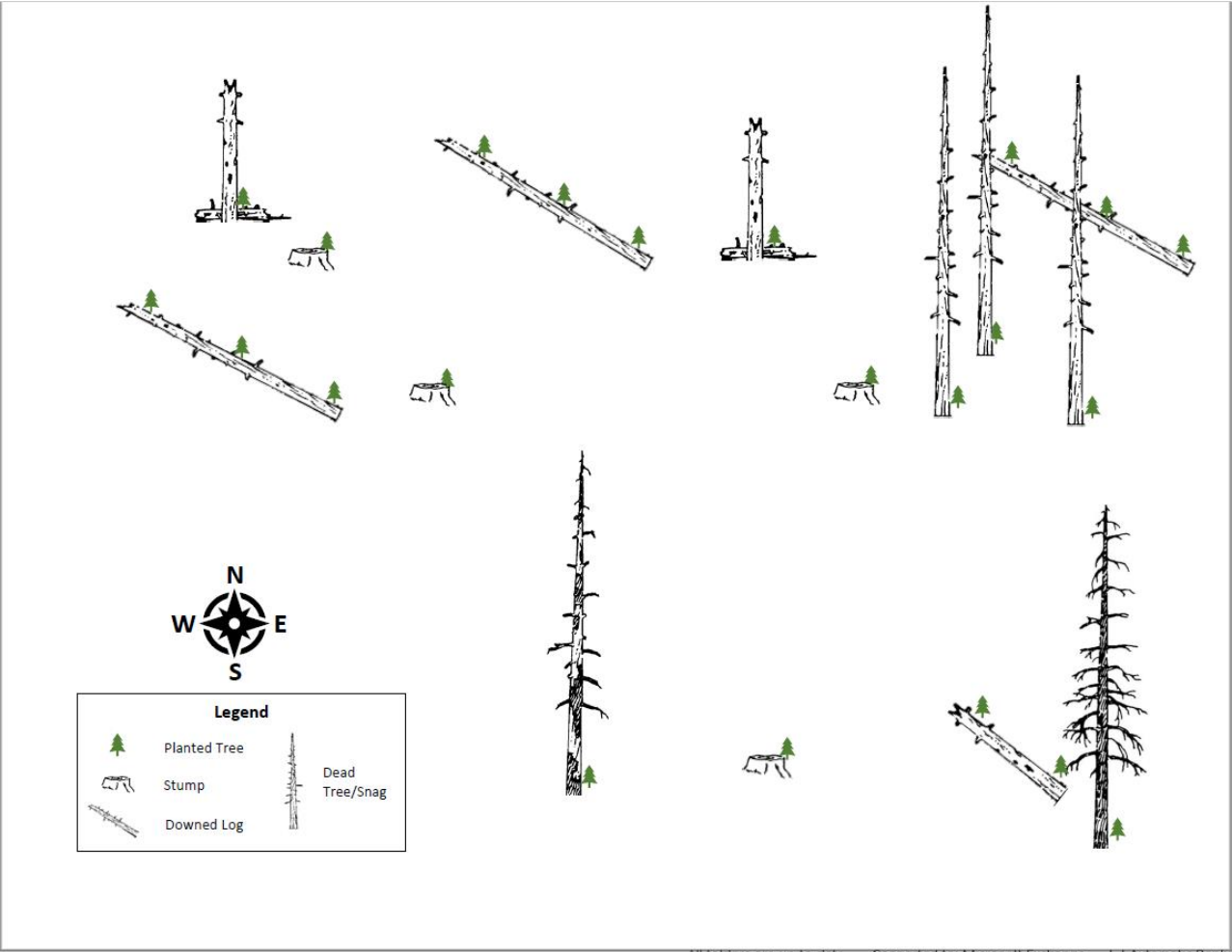


Exhibit 09 - Planting Inspection Sheet

PLANTING INSPECTION SHEET (R-1 FS-2470-9)

[illegible]

Calculations:

Submitted by: _____ Date: _____

Exhibit 10 – Planting Inspection Instructions

Instructions (Side 2 of Planting Inspection Sheet)

1. Locate and mark the plot center on the ground.
2. Locate and examine planted trees for above ground criteria. Record results in top row of Column 2 of Inspection Sheet.

Above Ground

✓	Satisfactory tree	C	Scalping, clearing, or planting spot preparation violation
S	Spacing violation	W	Wrong species
P	Planting spot selection violation	T	Cull tree
X	Shade violation or protection violation	H	Auxiliary/Attempted planting hole
D	Planting depth or exposed root violation	Z	Missed planting hole violation (Auger Planting)
A	Stem position violation (improper angle) or stem damage	E	Existing acceptable seedling/tree violation
F	Firmness violation (tree improperly tamped or loose)		

Commented [EJ3]: This and the next code are not included in R1 handbook example. Used in R4 planting contracts. Do you want to keep? Add other codes?

3. Determine from Table 1 the average number of desired plantable spots for the plot based on the specified average spacing. Record results in Column 3.
4. Determine and record in Column 4 the number of spots void of planted trees that are unplantable due to acceptable existing natural regeneration or on-the-ground conditions.
5. Subtract Column 4 from the average number of desired planting spots (Column 3) and record the difference as the actual number of plantable spots on which trees should be planted on the plot in Column 5.
6. Determine the maximum number of allowable trees from Table 2 and record in Column 6.
7. Record the number of trees planted on the plot from Column 2 in Column 7.
8. Determine wasted trees by subtracting the maximum number of allowable trees (Column 6) from those planted (Column 7) and if more than zero, record in Column 8.
9. Record the number of planted trees meeting the above-ground contract specifications from Column 2 in Column 9. The maximum number of satisfactory trees to be credited shall not exceed the maximum number allowable recorded in Column 6.
10. Determine and dig at least the minimum number of trees from those determined satisfactory above ground in accordance with Table 3 and record in Column 10. Dig trees starting with those closest to the plot center and progressing outwards.
11. Tabulate below-ground results in the bottom row of Column 2 and record the number of trees meeting below-ground contract specifications in Column 11.

Below Ground

✓	Satisfactory tree	F	Firmness violation (air pockets, tree improperly tamped, loose)
R	Root configuration violation	L	Altered root length violation or Damaged roots
M	Foreign material in planting hole	O	Planting hole orientation violation

12. After all plots have been taken and recorded, record the sum of each column at bottom of sheet and calculate the planting quality using the following formula. Record the final quality percentage, and complete the form with your signature, title, and date of final inspection results.

Planting Quality Percent

$$\frac{\text{Number of Satisfactory Trees Above Ground (Column 9)}}{\text{Number of Plantable Spots (Column 5)}} \times \frac{\text{Number of Satisfactory Trees Below Ground (Column 11)}}{\text{Number of Dug Trees (Column 10)}} \times 100$$

Tables: Portion of Relevant Tables from Planting Inspection Method. For complete Tables, see Planting Inspection Method.

Table 1: Avg. No. Plantable Spots

Spacing	1/50 th ac	1/100 th ac
10x10	9	4
11x11	7	4
12x12	6	3
13x13	5	3
14x14	4	2
15x15	4	2

Table 2: Max. Allowable Trees

# Plantable Spots	Max. Trees	# Plantable Spots	Max. Trees
1	2	6	7
2	3	7	8
3	4	8	10
4	5	9	11
5	6	10	12

Table 3: Min. Trees to Dig

No. of Satis. Trees	Min. No. of Trees to Dig
1	1
2-6	2
7-9	3

Exhibit 11 – Planting Inspection Method

Planting contracts require intensive inspection. The quality control plan shall be consistent with the following procedures. There are two phases of contract inspection: (1) observing tree handling, dipping procedures, wasting of trees, and planting technique while planting is in progress, and (2) plot inspection where the inspector checks quality of planting using systematically placed plots.

Inspecting for Planting Quality. The primary factor in determining the contractor's payment is the plot inspection. Use the following process and standard inspection form (R-1 FS-2470-9; *Exhibit 09*). Variations to this may be done only with the approval of the Contracting Officer's Representative (COR). The inspector shall sign the inspection form and initial any changes. All inspectors should fully understand the inspection forms and procedures for completing the inspection.

Suggested Equipment

- Clipboard, pencils, and permanent marker
- Inspection sheets
- Full contract (with exhibits) and task orders, or knowledge of unit requirements
- Fifty-foot logger's tape.
- Plot pole or shovel with swivel on handle to attach the tape.
- Flagging
- Screwdriver, ice-pick, or garden trowels that aid in below ground inspection
- Tile spade, hog nose spade, planting hoe for hoe planting
- Slope correction table (see *Plot Design* section below)
- Clinometer or Abney.

Plot Design

Install a series of 1/20th, 1/50th, or 1/100th-acre inspection plots, with plot size dependent on task order Detailed Information Chart specifications. In general, narrower spacings should use the 1/50th or 1/100th acre plot. Wider spacings should use the 1/20th or 1/50th acre plots.

Sampling intensity and plot distributions are described in *PWS Section 08*.

Plot radius will vary depending on slope. After determining slope percent and adjusting for slope using the slope correction table below, determine the plot radii to provide the necessary plot size. Use this radius for the full plot. Do not compensate by changing radii or raising or lowering the tape when going around the plot. Such adjustments have already been compensated for.

Slope Correction Table

Plot Size in Acres				Plot Size in Acres				Plot Size in Acres			
Slope %	1/100	1/50	1/20	Slope %	1/100	1/50	1/20	Slope %	1/100	1/50	1/20
0	11.8	16.7	26.3	56	12.6	17.8	28.2	88	13.6	19.2	30.4
5	11.8	16.7	26.3	58	12.7	17.9	28.4	90	13.7	19.3	30.6
10	11.8	16.7	26.5	60	12.7	18.0	28.5	92	13.7	19.4	30.7
15	11.8	16.7	26.5	62	12.8	18.1	28.6	94	13.8	19.5	30.8
20	11.9	16.8	26.6	64	12.8	18.1	28.7	96	13.9	19.6	31.0
25	12.0	16.9	26.7	66	12.9	18.2	28.8	98	13.9	19.7	31.2
30	12.1	17.0	26.9	68	12.9	18.3	29.0	100	14.0	19.8	31.3
35	12.1	17.1	27.1	70	13.0	18.4	29.1	102	14.1	19.9	31.5
40	12.2	17.3	27.4	72	13.1	18.6	29.2	104	14.1	20.0	31.6
42	12.3	17.3	27.4	74	13.1	18.6	29.3	106	14.2	20.1	31.8
44	12.3	17.4	27.5	76	13.2	18.7	29.6	108	14.3	20.2	31.9
46	12.4	17.5	27.6	78	13.3	18.8	29.7	110	14.4	20.3	32.1

Slope Correction Table

Plot Size in Acres				Plot Size in Acres				Plot Size in Acres			
Slope %	1/100	1/50	1/20	Slope %	1/100	1/50	1/20	Slope %	1/100	1/50	1/20
48	12.4	17.5	27.7	80	13.3	18.8	29.8	112	14.4	20.4	32.2
50	12.5	17.6	27.9	82	13.4	18.9	29.9	114	14.5	20.5	32.5
52	12.5	17.7	28.0	84	13.5	19.1	30.1	116	14.6	20.6	32.6
54	12.6	17.8	28.1	86	13.5	19.1	30.3	118	14.6	20.7	32.7

Above Ground Inspection

Inspect and record the **above ground** condition of each tree. Working in a clockwise direction from true north, locate, examine, and record the condition of planted trees in spaces under Column 2 of the inspection form. Use codes listed below. A poorly planted tree may have more than one violation; however, only one code may be listed. Identify the most severe. Refer to *PWS Section 4.4* for reference specifications.

- ✓ Satisfactory tree above ground.
- **S - Spacing violation.** A tree that has been planted closer than one-half the distance to the next plantable spot is a violation unless otherwise stated in the contract or task order. For example, if the spacing is 10 feet by 10 feet and there is a 50% variance to select the best planting spot, and a planted tree is closer than 5 feet to another planted tree, one tree is in violation for spacing. If one of the trees is improperly planted due to another reason, charge the spacing violation to the improperly planted tree and check the remaining good tree as properly planted.
- **P - Planting spot violation.** Tree planted in debris, loose soil, duff, ashes, or similar material. Improper planting spot selection based on microsite or shade preference.
- **X - Shade or protection violation.** Shade or seedling protection is not consistent with specifications for the unit.
- **D - Planting depth or exposed root violation.** Tree planted too deep or shallow.
- **A - Stem position (improper angle) or stem damage violation.** The stem should be oriented between vertical and 90 degrees with the slope plane. Improper angle may result from improper hole opening with hoes. If the tree looks erect above ground but is slanted below ground, then a below ground violation instead should be cited. Any planter-caused stem damage that could cause tree mortality should be recorded here.
- **F - Firmness violation.** Trees should be tamped as firmly as soil conditions allow. Trees should not pull easily from soil. The inspector may gently tug on the stem. If tree comes up to expose roots (below the root collar), then the tree was not firmed up and is in violation. This test must be used with caution in very light or sandy soils.
- **C - Scalping, clearing, or planting spot preparation violation.** Scalp or cleared area is too small or too shallow. Spot was not properly prepared.
- **W - Wrong (incorrect) species.** Species planted in area of unit where it is not supposed to be.
- **T - Cull trees.** Cull tree is planted. Use when the contract specifies that the contractor shall not plant trees that do not meet specific size ranges or tree quality.
- **H - Auxiliary/Attempted hole violation.** Additional hole or auxiliary hole is not filled in or collapsed.
- **Z - Missed planting hole violation (Auger Planting).** Tree not planted in plantable spots when a satisfactory hole was dug.
- **E - Existing acceptable seedling/tree violation.** Tree planted too close to existing acceptable seedlings or trees.

Determine the average number of planting spots from Table 1 and record results in Column 3 of the inspection form.

Table 1 – Average Number of Plantable Spots

Spacing	1/20 th Acre Plot	1/50 th Acre Plot	1/100 th Acre Plot
5 x 5	87	35	17
6 x 6	61	24	12
7 x 7	44	18	9
8 x 8	34	14	7
9 x 9	27	11	5
10 x 10	22	9	4
11 x 11	18	7	4
12 x 12	15	6	3
13 x 13	13	5	3
14 x 14	11	4	2
15 x 15	10	4	2
16 x 16	9	3	2
17 x 17	8	3	2
18 x 18	7	3	1
19 x 19	6	2	1
20 x 20	5	2	1

Determine the number of unplantable spots. After determining the average number of planting spots on the plot, scan the plot for areas void of planted trees. Look for voids the planters missed. Then check the voids to see if it is an unplantable spot due to ground conditions or acceptable established seedlings or trees. Record the number of unplantable spots in Column 4.

An unplantable spot as defined by the contract is an area within the specified spacing limits in which it is not possible to plant a tree according to specifications, and no tree has been planted (refer to *PWS Sections 2.0 – Definitions and 4.4.2*). For example, if a single unplantable area of 64 square feet exists on an 8-by 8 foot spacing, one unplantable spot is recognized. When a tree has been planted, the contractor, by default, has determined it to be a plantable spot. In terms of the inspection, the spot with a planted tree is a plantable spot no matter what the inspector finds after the fact.

Determine the number of planting spots by subtracting Column 4 from the average number of planting spots (Column 3). Record the result in Column 5 of the inspection form.

Determine the maximum number of allowable trees from Table 2, and record in Column 6 of the inspection form.

Table 2 - Maximum Allowable Trees Based on Number of Plantable Spots (as determined from Table 1)

Plantable Spots	Maximum Trees	Plantable Spots	Maximum Trees	Plantable Spots	Maximum Trees
1	2	9	11	16	19
2	3	10	12	17	20
3	4	11	13	18	22
4	5	12	14	19	23
5	6	13	16	20	24
6	7	14	17	21	25
7	8	15	18	22	26
8	10				

Record the number of trees planted on the plot (also listed in Column 2) in Column 7 of the inspection form.

Determine wasted trees by subtracting the maximum number of allowable trees in Column 6 from those planted (Column 7). If more than 0, record in Column 8.

Record in Column 9 the number of planted trees meeting above ground specifications from Column 2, only up to the maximum listed in Column 6 of the inspection form.

Below Ground Inspection

Determine the minimum number of trees to be inspected below ground utilizing Table 3, based on number of trees that are satisfactory above ground (Column 9). Record the result in Column 10 of the inspection form. To avoid bias, dig trees nearest plot center first and progress outward. Do not dig any trees that were unsatisfactory in the above ground inspection. Correctly replant sampled trees immediately. Use moist mineral soil to pack roots.

Table 3 - Minimum Number of Trees to Dig

Number of Above-Ground Satis. Planted Trees (Column 9)	Minimum Number of Trees to Dig*
1	1
2-6	2
7-9	3
10-12	4
13-16	5
17 +	6

**Note: This is a minimum. The inspector may dig more trees than the minimum needed per plot.*

Inspect and record the below-ground condition of each planted tree in spaces in bottom row of Column 2 on the inspection form using appropriate codes listed below. Record the number of trees meeting below-ground specification in Column 11. Refer to *PWS Section 4.4* for reference specifications.

✓ **Satisfactory** Tree Below Ground.

- **R - Root Configuration Violation.** Root systems must not be twisted, jammed in one plane, or curved in the shape of the letters U, J, or L. Individual lateral roots may be slightly bent like the letters J or U but the primary vertical root system cannot be distorted. Container plug must not be jammed from the top (accordion effect) or the side (flattened).
To inspect for this violation, dig a rectangular shaped hole on one side of the tree. Start the hole far enough away from the tree stem (5 - 10 inches away) so that roots are not disturbed in the process of inserting the spade. This is best done with a tool like a tile spade. Once the primary hole is dug, probe toward the root system with a pointed instrument such as a screwdriver, ice pick, or similar tool to explore the seedling roots for orientation.
- **M - Foreign Material in Planting Hole Violation.** Holes must not contain large rocks, sticks, litter, cones, or other foreign debris. Inspect the same as described for root configuration. In hoe planting, if rocks, roots, and pieces of wood are present in soil prior to opening the hole, they are not considered foreign material to the hole.
- **F - Firmness Violation.** Firmness, as determined from below ground, is done while probing the root system as described under root configuration. Soil should be nearly as firm as the undisturbed surrounding soil. There should be no air pockets where the soil is not firm. Firmness may be a problem in the bottom of auger holes if planters have not firmed soil progressively upward.
- **L - Altered Root Length Violation or Damaged Roots.** If the dug tree has an obviously shortened root system, consider it as an improperly planted tree below ground. The contract shall state a minimum root length; trees with substandard roots should not be planted and considered a violation if they are planted.

Fresh root cuts can often be distinguished from roots cut at the nursery or during tree preparation. Living inner tissue of roots cut in advance of planting should be brown at the cut. The brown color may extend up the root under the bark for a short distance. Freshly cut roots will be white under the bark unless roots are dead when cut. Root shortening violations can also be detected during inspection while work is in progress.

- O - Planting Hole Orientation Violation. This violation is seldom observed in absence of above ground stem angle or root configuration violations. Occasionally, trees with small root systems can be propped up following slit planting. Roots may not be distorted and angle looks okay from above ground, but hole is not properly opened. Examples would be a V-shaped hole or a hole not vertical with slope plane.

Planting Quality Percent

After all plots have been taken and recorded for the pay or sub item, calculate the planting quality using the following formula:

$$\frac{\text{Number of Satisfactory Trees Above Ground (Column 9)}}{\text{Number of Plantable Spots (Column 5)}} \times \frac{\text{Number of Satisfactory Trees Below Ground (Column 11)}}{\text{Number of Dug Trees (Column 10)}} \times 100$$

Record the final planting quality percent on the inspection form and sign with signature, title, and date of the final inspection results.

Exhibit 12 - Sample Planting Inspection and Explanatory Notes

Sample Planting Inspection Sheet

FOREST Kootenai		DISTRICT Cascade		INSPECTOR R. Needles		CONTRACT/TASK ORDER NUMBERS 1284LL21D0001/12363N24F4059																
COR P. Greenup		PLOT SIZE 1/50 th ac		STAND NUMBER 20003001		CONTRACTOR TreesRUs, Inc		SUB ITEM NUMBER 3.1														
ACRES 20		AVERAGE SPACING 12x12		MAX TREES/ PLOT 7				SHEET NUMBER 1 OF 1														
DATE	Plot No.	(2) Examine & Tally Planted Trees*											Desired No. of Plantable Spots	No. Unplantable Spots	Actual No. Plantable Spots	Maximum No. Allowable Trees	No. Planted Trees	No. Wasted Trees	No. Trees Status Above Ground	No. Trees Dug	No. Trees Status Below Ground	Remarks
		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)										
5/1	1	C	✓	✓	✓	✓	✓						6	0	6	7	6	0	5			Small scalp
			✓		✓															2	2	
	2	✓	✓	✓	✓	✓	✓	✓					6	1	5	6	7	1	6			Unplantable: Rock
		✓			M															2	1	Rotten wood in hole
	3	✓	X	✓	✓	✓							6	0	6	7	5	0	4			Did not shade by stump
				✓	✓															2	2	
	4	✓	✓	✓	✓	✓	S	✓					6	2	4	5	7	2	5			Seedlings too close, natural WL
		✓	✓																	2	2	
	5	✓	✓	✓	✓	✓	✓	✓	✓				6	0	6	7	7	0	7			
		✓			F		✓													3	2	Air pocket
	6	✓	✓	X	✓	✓	D						6	0	6	7	6	0	4			No shade; shallow planted tree
		✓			✓															2	2	
	7	✓	✓	✓	✓	✓	✓	✓					6	0	6	7	7	0	7			
		✓		✓	✓	✓														3	3	
	8	✓	✓	✓	✓	✓	✓	✓					6	1	5	6	7	1	6			Standing water
		✓				✓														2	2	
	9	✓	✓	✓	✓	✓	✓	✓	✓	✓			6	0	6	7	8	1	7			
				✓		L	✓													3	2	Roots were trimmed
10		✓	✓	✓	✓								6	3	3	4	4	0	4			Acceptable established trees on plot
		✓			✓															2	2	
*First line for above ground. Second line for below ground.		TOTALS											60	7	53	63	64	5	55	23	20	FINAL % = 90%
Calculation: $\frac{\text{Col. 9}}{\text{Col. 5}} \times \frac{\text{Col. 11}}{\text{Col. 10}} \times 100 = \frac{55}{53} \times \frac{20}{23} \times 100 = 90.235$ Wasted trees on plots = 5 Submitted by Rob Needles, Inspector. 5/2/2003 All information is correct and complete.																						

SAMPLE PLANTING INSPECTION SHEET EXPLANATORY NOTES

Note that the Contractor filled out the Inspection Sheet header completely. The Forest, District, Contract/Task Order Numbers, COR, Stand Number, Sub Item, Acres, and Avg. spacing are information provided by the Government either directly or within the contract.

The Contractor provided the name of the (Contractor) Inspector, plot size, max trees/plot (based on contract tables), sheet number, and calculated the final quality. The Contractor assured that all columns and math were correctly done. The Contractor's inspector signed and dated the inspection records and certified that the information was correct and accurate.

Based on the sample Inspection Sheet, the plot information was completed based on these observations:

Plot 1: Six trees were planted on the plot. The first tree did not have the required scalp width and received a C for scalping/clearing violation. Other trees were satisfactory above ground. There were no voids on the plot greater than 144 sq. feet (12 x 12 spacing), so "0" unplantable spots are recorded. The inspector dug two trees, which is the minimum number based on 5 satisfactory trees in the above ground inspection. Both were properly planted.

Plot 2: There were seven planted trees on the plot, all satisfactory above ground. The inspector found a void on the plot with no trees that was larger than 144 sq. ft. The inspector checked for plantability and it was unplantable due to rock. One unplantable spot was recorded. This caused one wasted tree in column 8. Note that the satisfactory trees recorded in column 9 is six (not seven- cannot exceed the maximum allowable trees, col. 6). In the below ground inspection, a piece of rotten wood was found in the hole, which is a foreign material violation (F).

Plot 3: There were five planted trees. One tree was not planted by a stump, which was a shade violation (X) since the contract required that all trees be planted by stationary shade when it is available. There were no voids on the plot of 144 sq. ft that were unplantable. The two sample trees that were dug were satisfactorily planted based on below ground requirements.

Plot 4: There were seven planted trees. Two trees were planted five feet apart (minimum spacing was 50% of 12 feet or 6 ft). A spacing violation (S) is recorded for one of the trees. The other trees were satisfactory. The inspector found two voids on the plot that were unplantable due the presence of acceptable established WL seedlings, each at least 144 sq. ft (approximate square or circle) and recorded two unplantable spots in column 4. Based on only four planting spots, the maximum allowable trees is five. Since seven trees were planted, two trees are wasted. Only five trees are recorded in column 9. Two trees were dug and both were properly planted.

Plot 5: Seven trees were planted and all were satisfactorily planted above ground. There were no unplantable voids equal to 144 sq. ft. The below ground inspection revealed an air pocket in the lower part of the hole which is a Firmness violation (F).

Plot 6: There were six trees planted. One tree was not shaded which is a shade violation (X) since the contract required shade when it was available. Another tree was planted high with the root collar above the ground surface so a depth violation (D) was recorded. Four trees are identified as satisfactory. Based on this, two trees were dug from the four that were satisfactory above ground; and found to be planted properly.

Plot 7: There were seven trees planted all of which were satisfactory. No unplantable voids existed. Three trees were dug based on seven satisfactory trees; they were satisfactory.

Plot 8: Seven trees were planted, all of which were satisfactory. Although they tended to be closer than 12 feet apart, they were not closer than 6 feet apart, so no violation occurred. There was a void on the plot about 14x14 in size with no trees and was found to be unplantable due to standing water. One unplantable spot was noted. Since seven trees were planted but the maximum number of allowable trees is six (due to one unplantable spot), there is one wasted tree in column 8. Only six trees count as satisfactorily planted above ground (cannot exceed col. 6), thus two trees were dug and were satisfactory.

Plot 9: Eight trees were planted. They tended to be closer than 12 feet apart but were not closer than 6 feet, thus no violation. There were no unplantable voids on the plot. The maximum allowable trees is seven, thus one wasted tree is recorded in col. 8. Three trees were dug based on seven satisfactory planted trees (cannot exceed col. 6). The roots on one of the trees had

been stripped either during planting or before planting, (if before planting, the planter should not have planted the tree) resulting in a violation for altered root length (L).

Plot 10: Four trees were planted, and each was satisfactorily planted. Almost half of the plot could not be planted because there were small trees that met the requirements for acceptable established trees. This accounted for about three areas of 12x12 feet in size that were taken up with the residuals accounting for three unplantable spots. Two trees were dug and found to be satisfactory.

Following the inspection, all numbers were checked and totaled. The final quality is calculated based on the equation provided on the backside of the inspection sheet.

Evaluation: It should be noted that as a final inspection, the Contractor failed to meet the required 93% quality for acceptable work. He would receive a 10% reduction of the bid price for this unit or pay item according to *PWS Section 10.3*. Although the Contractor over-planted, it helped to compensate for the above ground errors. It was the below ground errors that caused work to be unacceptable.

The over-plant also resulted in 5 wasted trees on the plots, which computes to 25 trees per acre that are excess. The Contractor should be assessed the wasted tree charges.

Exhibit 13 – Planting Inspection Sheet, Spanish Version

HOJA DE INSPECCIÓN DE PLANTACIÓN – ESPAÑOL (R-1 FS-2470-9)

Commented [EW4]: Spanish version adapted using multiple internet resources. Would be nice to have it checked by someone fluent in Spanish.

[illegible]

Firmado Por: _____ Fecha: _____

Exhibit 14 – Planting Inspection Instructions, Spanish Version

HOJA DE INSPECCIÓN DE PLANTACIÓN – ESPAÑOL (Continuado)
Instrucciones (pagina 2 de la Hoja de inspección de plantación)

- 1. Localice y marque el centro del ploto en el suelo.
- 2. Localice y examine los árboles plantados para determinar los criterios sobre el suelo. Anote los resultados en la columna 2 de la hoja de inspección.

Sobre el Suelo			
√	Arbol satisfactorio	C	Violación del desbroce o la preparación del lugar de plantación
S	Violación del espacio	W	Especies equivocadas
P	Selección fallida del lugar de plantación o micrositio	T	Árbol de sacrificio: muerto, debil, pequeño
X	Violación de la sombra o violación de la protección	H	Agujero auxiliar
D	Violación de la profundidad de plantación	Z	Violación del hoyo de plantación omitido (Barrena)
A	Violación de la posición del tallo del árbol: ángulo incorrecto	E	Violación de árbol aceptable existente
F	Violación de firmeza: árbol suelto, árbol mal apisonado		

- 3. Determine a partir de la tabla 1 el número promedio de lugares plantables deseados para el ploto según el espaciamiento promedio especificado. Anote los resultados en la columna 3.
- 4. Determine y anote en la columna 4 el número de lugares sin árboles plantados que no se pueden plantar debido a una regeneración natural existente o a condiciones del terreno.
- 5. Reste la columna 4 del número promedio de lugares plantables y anote la diferencia en la columna 5. Este es el número real de lugares plantables donde se deben plantar árboles.
- 6. Determine el número máximo de árboles permitidos a partir de la tabla 2 y anote en la columna 6.
- 7. Anote el número de árboles plantados en el ploto de la columna 2 en la columna 7.
- 8. Determine los árboles desperdiciados: reste el número máximo de árboles permitidos en la columna 6 de los plantados (en la columna 7) y, si es mayor que cero, anote en la columna 8.
- 9. Anote el número de árboles plantados que cumplen con las especificaciones del contrato para sobre el suelo de la columna 2 en la columna 9. La cantidad máxima de árboles satisfactorios que se acreditarán no excederá la cantidad máxima registrada en la columna 6.
- 10. Determine y excave al menos el número mínimo de árboles satisfactorios sobre el suelo a partir de la tabla 3 y anote en la columna 10. Los árboles se excavarán comenzando por los más cercanos al centro del ploto y avanzando hacia afuera.
- 11. Marque los resultados subterráneos en la columna 2 y anote el número de árboles que cumplen con las especificaciones de contacto para subterráneos en la columna 11.

Bajo Tierra			
√	Satisfactorio	F	Firmeza: bolsas de aire, árbol mal apisonado, suelto
R	Violación de la configuración de las raíces de árbol	L	Violación de la longitud de la raíces de árbol, alterada
M	Material extraño en el agujero de plantación	O	Violación de la orientación del agujero de plantación

- 12. Después de completar y registrar todas los plotos, registre la suma de cada columna en la parte inferior de la hoja y calcule la calidad de plantado usando la siguiente fórmula. Registre el porcentaje de calidad final, su firma, su título y fecha de los resultados de la inspección final.

Porcentaje de Calidad de Plantado

Número de árboles plantados satisfactorios sobre el suelo (de la columna 9)

Número real de lugares plantables donde se deben plantar árboles (de la columna 10)

X

Número de árboles bajo tierra satisfactorios (de la columna 11)

Número de árboles excavados (de la columna 5)

X

100

=

Porcentaje de Calidad (%)

Tablas: Porción de tablas relevantes del Método de Inspección de Plantacion. Consulte Método de Inspección de Plantación para tablas completas.

Tabla 1: promedio de lugares plantables			Tabla 2: máximo de árboles permitidos				Tabla 3: mínimo de árboles para excavar	
Espaciado	1/50 th ac	1/100 th ac	lugares plantables	árboles máximos	lugares plantables	árboles máximos	árboles satisfactorio	árboles para excavar
10x10	9	4	1	2	6	7	1	1
11x11	7	4	2	3	7	8	2-6	2
12x12	6	3	3	4	8	10	7-9	3
13x13	5	3	4	5	9	11		
14x14	4	2	5	6	10	12		
15x15	4	2						

REFORESTATION ACTIVITY RECORD

COMPARTMENT		STAND	ACRES	PROJECT MAP
SUB COMPARTMENT		SALE NAME		
Site data	ASPECT			
	%SLOPE			
	ELEVATION			
	HABITAT TYPE			
ORIGIN CC SW ST S REGENERATION METHOD YEAR INITATED				
REFORESTATION FINANCED KV P&M				
SITE PREPARATION: METHOD AND FINANCING KV P&M BD				
DATE		PROJECT DESCRIPTION		
The unit was planted by _____ with _____ container, _____ bareroot, Lot _____ at _____ TPA _____ container, _____ bareroot, Lot _____ at _____ TPA _____ container, _____ bareroot, Lot _____ at _____ TPA				

STAND NUMBER

SALE NAME

Stand Number _____ Dates Planted _____
Trees Planted/ Day _____

Bag Temperature Morning: _____ Midday: _____

Overall weather conditions:

Satisfactory stocked

Unsatisfactory stocked

Total acres	Total acres	Total acres	
Sat. stocked _____	Unsat. stocked _____	Non-plantable _____	Total _____

Reforestation Project Complete

Date _____ Signature _____ Title _____